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Projekt dyplomowy

Automated calibration of synchrotron beamline for micro-XRF
experiments

Zautomatyzowana kalibracja linii synchrotronowej dla
eksperymentów micro-XRF

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1 Application purpose and design assumptions

The Micro-XRF App was designed as a dedicated tool to support work on the PolyX beamline, intended for comprehensive management of calibration procedures in X-ray fluorescence analysis. The software was implemented using MATLAB App Designer environment, allowing direct integration with the beamline control system while providing an intuitive graphical user interface (GUI).

1.1 Measurement of the standards

The primary purpose of the application is automation of the measurement process of XRF calibration standards. This measurement process is essential for determining the calibration parameters and detector characteristics. The application was specifically designed to work with a dedicated sample holder, where the individual standards are arranged in a fixed layout.

1.2 Simplifying the measurement procedure

One of the main goals was to create a user-friendly tool accessible to operators or users who may not have detailed knowledge about low-level beamline control. The application achieves this through an intuitive and interactive GUI. The stages of the experiment, from selecting standards for the measurements, through data acquisition, to analysis, are conducted within a single application, without the need to switch between different programs or terminals.

1.3 Comprehensive calibration process

The application is designed to guide user through the entire calibration process in a smooth and structured manner. It enables the following steps to be carried out:

- **Efficient measurement:** Automation of the measurement sequence and acquisition of XRF maps for the selected standards, carried out within a single procedure, significantly reducing the time required for data collection.
- **Energy calibration:** Based on the automatically detected peaks.
- **Resolution calibration:** Determination of FWHM as a function of energy.
- **Sensitivity calibration:** Calculation of elemental sensitivity curves for individual spectral lines of the selected standards.

In summary, the developed application is designed to transform the multi-step procedure of measuring XRF standards and performing the corresponding calibration into a simple, fast and error-resistant process.

1.4 Dedicated standard holder and sample layout

The foundation of the automated measurement procedure is a dedicated sample holder, designed specifically for calibration on the PolyX beamline. The assembly consists of X-ray fluorescence standards manufactured by MicroMatter™, deposited on a thin Mylar®polyester film (type M6.3, thickness 6.3 μm). The holder accommodates 30 standard samples arranged in a regular 5×6 grid, covering a wide range of atomic numbers from sodium ($Z = 11$) to lead ($Z = 82$).

The actual appearance of the holder, together with the mounted samples, is shown in Figure 1.



Figure 1: Photograph of the dedicated sample holder containing 30 MicroMatter standards.

The exact elemental composition and mass per unit area of each standard are provided in Table 1.

Table 1: Detailed composition of the calibration standards on holder. The values in brackets indicate the mass per unit area [$\mu\text{g cm}^{-2}$]. Elements marked in **bold** indicate those that are explicitly selected and analyzed by the software for the given sample.

Na (3.1) Cl (4.7) NaCl (7.8)	Mg (2.9) MgF ₂ (7.4)	Al (5.6)	Si (2.86) SiO (4.49)	Ga (3.3) P (2.79) GaP (45.5)	In (16.9) S (4.2) InSx (21.1)
K (10.9) Cl (9.9) KCl (20.8)	Ca (28.5) CaF ₂ (55.5)	Ti (57.1)	V (54.8)	Cr (51.9)	Mn (55.3)
Fe (53.2)	Co (49.6)	Ni (46.8)	Cu (49.3)	Zn (16.4) Te (32) ZnTe (48.4)	Ga (20.9) As (24.6) GaAs (45.5)
Se (47.5)	Cs (30.1) Br (18) CsBr (48.1)	Rb (20.6) I (30.6) RbI (51.2)	Y (26.9) YF ₃ (44.2)	Rh (48.5)	Ce (38.9) CeF ₃ (54.7)
Sm (34.8) SmF ₃ (48)	Dy (38) DyF ₃ (51.3)	Yb (38.2) YbF ₃ (50.8)	W (40.9) WO ₃ (51.6)	Au (57.2)	Pb (48.1)

1.5 Experimental setup procedure and application workflow

Effective automation of the experimental workflow nevertheless requires preliminary physical alignment of the measurement setup. The procedure begins by mounting the holder in an aluminum frame and positioning it within the PolyX beamline experimental hutch. Once the experimental setup is established, the user employs the beamline control software and the optical camera view to drive the sample stage motors in order to put a randomly selected reference sample directly into the beam path. This physical location serves as a reference point, which must be marked in the developed application to synchronize the coordinate systems. Once this alignment is complete, the user can select the desired standards in the program’s GUI and initiate the automated measurement routine.

The application’s internal logic was designed to synchronize coordinate-based sample positioning with subsequent spectral analysis and calibration. To provide an overview of the software’s operational logic, a comprehensive flowchart is presented in Figure 2. The diagram illustrates the workflow split into two main operational modes.

The **Beamline mode** begins by identifying standards that are available for measurement at a current beam energy and registering the reference coordinates. This is followed by the **Automated measurement loop**, which executes the spectrum acquisition and analysis to perform energy, FWHM and sensitivity calibrations.

Alternatively, the application features a distinct **Offline mode**. This mode operates independently of the beamline hardware, allowing the user to load previously measured datasets and perform post-experimental analysis.

The following sections will describe each of these stages in detail, illustrating their realization in the graphical user interface and explaining the corresponding algorithms and methodology.

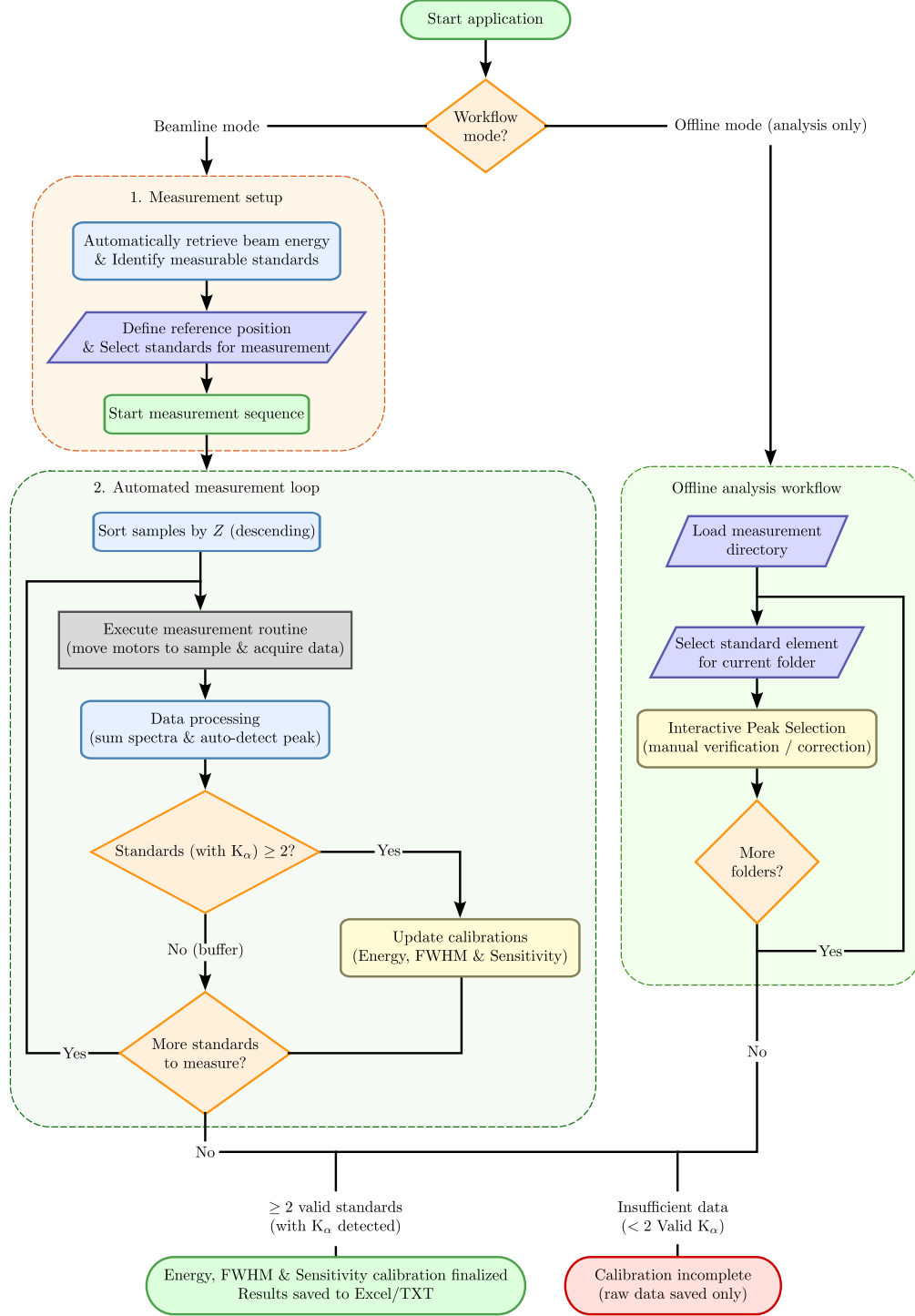


Figure 2: Flowchart illustrating the overall application workflow, distinguishing between the automated beamline mode, which enables measurement execution and calibration, and the offline analysis mode.

2 Software architecture

2.1 Interface structure

The “Set Calibration” tab (Figure 3) is an interface for setting up a measurement parameters, selection of standards for measurement of spectra and execution of calibration procedures.

The left side panel is a control section that displays the beam energy currently set on the PolyX beamline. This value synchronizes automatically based on the beamline configuration, but it can also be manually adjusted for corrections.



Figure 3: The “Set Calibration” tab interface showing the control panel, detectors calibration parameters (GAIN, ZERO, NOISE, FANO) and the calibration standards selection table with available characteristic lines of elements.

Below, the user can click the “Chose starting position” button to open a new window (Figure 4) with a graphical diagram of sample holder. Clicking on the reference sample in the popup window automatically captures the current motor coordinates corresponding to the targeted sample center and fills in the x [mm] and z [mm] values. There is also an option to manually adjust them if precise correction of the sample center is needed.

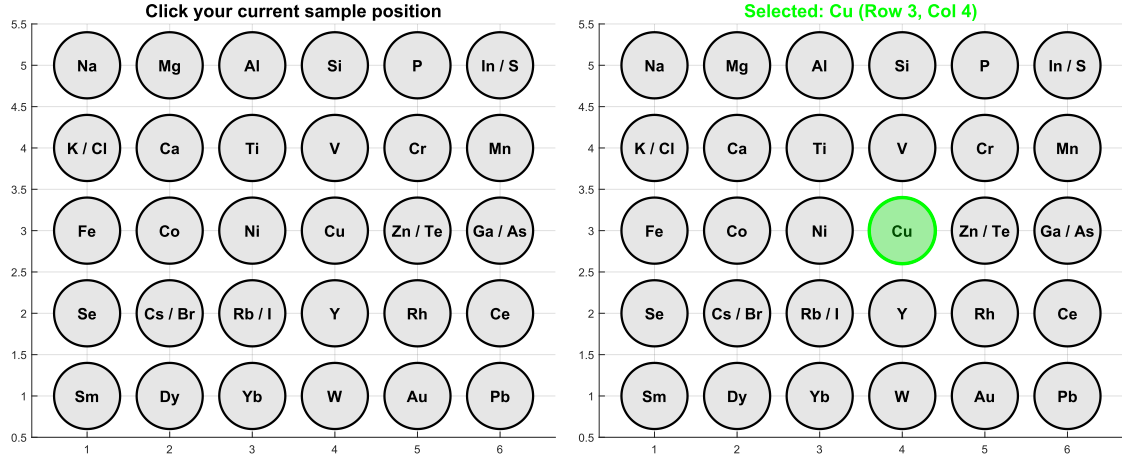


Figure 4: Sample position selection interface with calibration standards. A view of all the available standards (left) and the confirmed Cu position in row 3, column 4 highlighted in green (right).

The main part of the tab is the “Available Standards” panel which is an interactive table of all standards that can be measured on the beamline, used to choose elements for data acquisition. The elements are arranged in ascending order of atomic number, each with a large button and a set of small buttons representing emission lines: K_{α} , K_{β} , L_{α} , L_{β} , M. Clicking the large toggle button adds the selected element to the measurement list together with all emission lines that can be excited for this element. Clicking one of the small toggle buttons adds only that specific emission line. In both cases, the program will acquire a single XRF map for the chosen element, and the indicator light shows whether the element (and its selected lines) is currently included in the measurement. The application automatically checks the beam energy value. If the beam energy is too low to excite a specific emission line (for example, the K lines of Cu, Zn or Y in Figure 5), the selection buttons for that element will remain inactive (grayed-out). To quickly select a particular line of all elements that can be excited at a given energy, the user can also select an appropriate checkbox (for example ☒ K_{α}) in the upper header of the table.

Co	Ni	Cu	Zn
K_{α} ☒	K_{α} ☒	K_{α} ☐	K_{α} ☐
K_{β} ☒	K_{β} ☒	K_{β} ☐	K_{β} ☐
			L_{α} ☒
Y	Rh	In	Te
K_{α} ☐	L_{α} ☒	L_{α} ☒	L_{α} ☒
K_{β} ☐	L_{β} ☒	L_{β} ☒	L_{β} ☒
L_{α} ☒			

Figure 5: Part of the selection table of characteristic lines different elements at beam energy of 8970 eV. The program will acquire maps for elements with the lamp enabled (Co, Ni, Y, Rh, Te). Grayed-out buttons next to emission lines indicate that the energy is too low to excite them. A fully grayed-out Cu button means that no calibration lines are available for that element.

The top panels of the tab displays the energy: GAIN, ZERO, and the FWHM calibration parameters: NOISE, FANO. Upon startup, the program loads a proven set of parameters, optimized for the standard measurement geometry used on the beamline. These parameters are automatically overwritten after a new calibration, but the user can also edit them manually, and the program will remember these changes for the current session.

Once all desired standards have been selected, the left side panel provides a drop-down menu for choosing the ROIs width (σ ; equation 6) around the peak to be analyzed as the element signal.

Available options include $\pm 0.5 \cdot \sigma$, $\pm \sigma$, $\pm 2 \cdot \sigma$, and $\pm 3 \cdot \sigma$.

After everything is configured, clicking the “Start measurement” button initiates the measurement sequence. XRF maps are acquired and all the data will be saved for every selected element (emission line). After each subsequent element in the sequence processed, the energy calibration, FWHM calibration, and sensitivity calibration are updated accordingly, which also refreshes the parameter values displayed at the top of this tab and is additionally visualized in the interface as shown in Figure 6.



Figure 6: Visualization of measurement status in the standard selection panel. A green font color of the element symbols indicates that the measurement sequence has been successfully completed.

The application is designed to give the user full control over the input data used in the calibration algorithms. After the measurement is completed (that is, when the element name is shown in green), the spectral data are stored in the program’s memory. At any moment, the user can decide which of these measurements should be included in the calibration and which should be excluded (for example, due to the measurement error or poor signal to noise ratio) simply by enabling or disabling the button for the corresponding line.

The “Update Sensitivity calibration” button recalculates the sensitivity parameters and updates all plots for the active lines.

The “Update Energy & FWHM calibration” button checks all currently enabled K_{α} -line buttons, on the basis of which both calibrations are performed, and then recalculates and updates all corresponding parameters and plots.

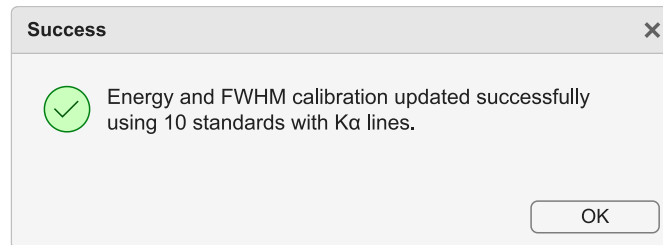


Figure 7: Message window notifying the user of the successful recalculation of the energy and FWHM calibration after clicking the “Update Energy & FWHM calibration” button.

2.2 Measurement data structures

To process data, the program uses entire folders (or directories) containing set of `.mat` files. A typical measurement sequence includes:

1. A header file terminated with `_HEADER`, which contains metadata for the measurement such as stepper motor configuration, preamplifier parameters, beam energy or monochromator type. An application loads this data into the structure `app.HeaderData`.
2. A snapshot file containing `_snapshot` in the name, loaded into `app.SnapshotData`.
3. Sequence files (for example, `...seq3_001.mat`, `...seq3_002.mat`), which are the most important from the program's perspective and contain spectral data. While loading the data, the program iterates through these files and sums their contents.

Inside the sequence files there are matrices that the program processes. The main matrix containing the counts is stored in the variable `dane1line` with dimensions [Number of detectors, Number of pixels, Number of Channels]. The program sums the matrix over all sequence files and then integrates over the pixels to obtain a single combined spectrum for each detector.

The `stats1line` matrix, with dimensions [2,4], contains the operating parameters of each detector for every measurement point. It includes the Real Time, Live Time, ICR (Input Count Rate) and OCR (Output Count Rate) values. It is also used to calculate the detector Dead Time (DT) using the formula:

$$DT = \left(1 - \frac{OCR}{ICR}\right) \cdot 100\%. \quad (1)$$

In the last sequence file, the complete intensity map of the primary beam and PIN-diode map for all pixels are stored. The program computes the average values $\overline{I_0}$ [V], which is used for determining the sensitivity, and $\overline{PIN}$ [V].

After all files are loaded, the program converts them into its own object structures in the application's memory. To analyze the data, the software integrates with the `xraylib`^[1] library as a reference for X-ray physics parameters. For example, for a titanium standard, the atomic number ($Z = 22$) is determined via the `SymbolToAtomicNumber('Ti')` function. Then, this Z value can be passed as an argument to retrieve the characteristic energies. For instance, the theoretical energy of the K_α line is obtained by executing the function `LineEnergy(22, KA_LINE)`, while the absorption edge is determined via `EdgeEnergy(22, K_SHELL)`.

These values are stored together with the experimental data and serve as a reference for energy calibration, peak identification and GUI feedback logic.

2.3 Integration with the PolyX code infrastructure

The application was designed to integrate with existing beamline infrastructure, enabling automated measurement workflows. It uses functions provided by the beamline software to communicate directly with hardware components such as motors and detectors. These functions, available on the beamline control computer, are accessible in MATLAB through proper environment configuration and path setup. Instead of re-implementing this functionality, the application calls the required routines when needed and integrates them into measurement workflow.

This integration allows complex measurement procedures to be carried out smoothly in an automated manner, without requiring the user to be familiar with the technical details of the beamline infrastructure.

2.4 Holder geometry and positioning

The physical geometry of the calibration standard holder corresponds to the layout shown in Figure 4. In this configuration, the samples are arranged in grid of 5 rows and 6 columns, with each sample represented as a 10 mm diameter circular spot. The distance between adjacent samples in both the horizontal and vertical directions is 17 mm. Based on this geometry, the application calculates the coordinates of successive samples.

In the application code, each position in the grid corresponds to a single sample, which may be either single-element or two-element. In the latter case, the position displays two element symbol separated by a “/” (for example, Zn/Te, Ga/As). The program recognizes this notation, splits it into individual symbols, and creates an internal mapping between each element and its corresponding sample position on the holder. This ensures that regardless of whether contains one or two elements, the application always knows its physical location.

After clicking Choose Starting Position button, the user selects a reference point that serves as the origin for the entire grid. The program reads the the current absolute motor positions using beamline infrastructure functions, storing these values as `init_x` and `init_z` (the coordinates of the reference sample center). From this starting point, the coordinates of the remaining samples are calculated as:

$$x = \text{init_x} - \text{spacing} \cdot \Delta\text{col}, \quad z = \text{init_z} - \text{spacing} \cdot \Delta\text{row}, \quad (2)$$

where:

- `spacing = 17 mm` – center-to-center distance between samples,
- `$\Delta\text{col}$ ,  $\Delta\text{row}$` – differences in column and row indices relative to the reference sample.

This setup allows the user to define only the reference point, while coordinates of all other samples are computed automatically.

2.5 Limit controls

The application uses software-defined stage axis limits configured directly on the beamline. After the reference position is set, and whenever the starting parameters are changed, a position availability verification mechanism is triggered, which updates the status of the limit indicators in the user interface. For each element, the software calculates the target (x, z) coordinates of the center of the corresponding sample based on the holder geometry and selected reference position, and then compares them with the limits returned by the controller for the X and Z axes.

If the calculated position is outside the allowed motion range, the application automatically blocks the corresponding standard: all line-selection buttons for that element are changed to black, the standard cannot be added to measurement queue, and any attempt to select it triggers a clear diagnostic message in the console, for example, in the case shown in Figure 8, “ $X = 20.00$ mm is outside limits for Cr.”

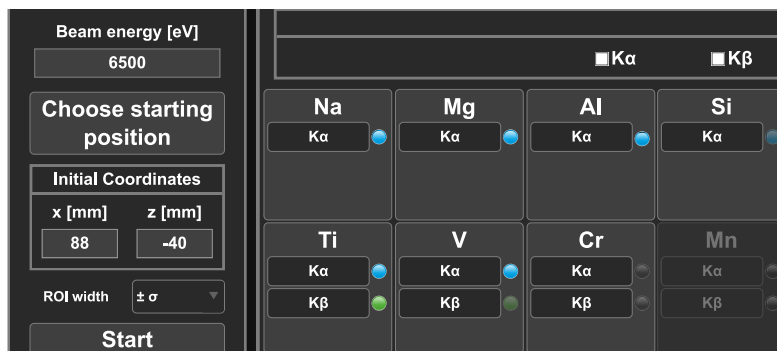


Figure 8: Visualization of the limit-checking mechanism in operation. With the reference position set on the sodium (Na) sample, the target coordinate calculated by the algorithm for chromium (Cr) exceeded the safe travel range of the stage. As a result, the application automatically disabled the selection of this element (inactive K_{α} and K_{β} line controls), and the indicator lamp changed to black.

These mechanisms prevents the user from accidentally scheduling a measurement at a physically unreachable position, directly improving the safety and reliability of the automated workflow.

2.6 Measurement sequence and real-time processing

Once the user confirms their selections and clicks the Start Measurement button, the application performs a series of background operations to optimize the measurement parameters and data acquisition process.

First, the software creates a list of “active” elements and reduces it to a list of active holder positions (slots). As discussed in section 2.4, each position corresponds to a single sample, which may be either single-element or two-element. Therefore, the program schedules one scan of XRF map per position rather than one scan per element. This approach avoids performing duplicate acquisition sequence on the same sample in case of two-element standards. Each XRF map is acquired over an area of $2 \times 2 \text{ mm}^2$, with a fixed step sizes in both directions of $X_{\text{step}} = Z_{\text{step}} = 0.5 \text{ mm}$. This results in a 5×5 grid, corresponding to a total of 25 measurement points per sample. It is worth noting that the entire analysis within the application is performed on a sum spectrum generated from all the measurement points.

The measurement order is then optimized by sorting the active positions so that the samples containing heavier elements (with higher atomic number Z) are measured first. This ordering improves the reliability of the results, as heavier elements typically produce stronger fluorescence signals, which enhances the effectiveness of the peak detection method.

For each measurement position, the application selects appropriate acquisition parameters. Within a given slot, the element with lower atomic number Z is identified, as it determines the scan time duration due to its generally weaker signal. The software then retrieves the monochromator type from the beamline infrastructure by calling the function $[\sim, \text{mt}] = \text{wmE}()$, where mt identifies a nickel (Ni), molybdenum (Mo), or silicon (Si) monochromator.

The acquisition time per point (Δt) is calculated to ensure sufficient counting statistics. The photoelectric absorption cross-section scales proportionally to Z^4 and the fluorescence yield (ω) for light elements increases approximately linearly with Z . The product of these two factors results in a signal intensity that scales as $\sim Z^5$. Consequently, to maintain comparable data quality, the exposure time must be extended in an inversely proportional manner (Z^{-5}) for lighter elements.

Taking this into account, and considering the adopted scan geometry (25 measurement points), the acquisition time per point is subsequently calculated according to the following rules:

- For Ni or Mo monochromator:

$$\Delta t = \begin{cases} 0.2, & Z \geq 30, \\ \frac{1.1 \cdot 10^8 \cdot Z^{-5}}{25}, & Z < 30. \end{cases} \quad (3)$$

- For Si monochromator:

$$\Delta t = \begin{cases} 2.0, & Z \geq 30, \\ \frac{1.1 \cdot 10^9 \cdot Z^{-5}}{25}, & Z < 30. \end{cases} \quad (4)$$

The selected value of Δt determines the stepper motor speed along the X-axis. Given the fixed step size, the motor movement speed is defined as:

$$V_x = \frac{X_{\text{step}}}{\Delta t}. \quad (5)$$

The XRF map is acquired using a predefined scan pattern symmetrically around the sample center, and the actual data acquisition is carried out by a dedicated beamline infrastructure script.

3 Data visualization

3.1 Spectrum display and plotting

The “View Data” tab (Figure 9) is used for visualization, qualitative verification, and analysis of the data acquired during the measurement.

The interface is organized into three main sections.

The central area of the tab features two plots corresponding to the two SDD detectors (Detector 1 and Detector 2). These plots present the sum spectra data gathered from the measurement sequence, with plot titles automatically indicating the currently displayed element.

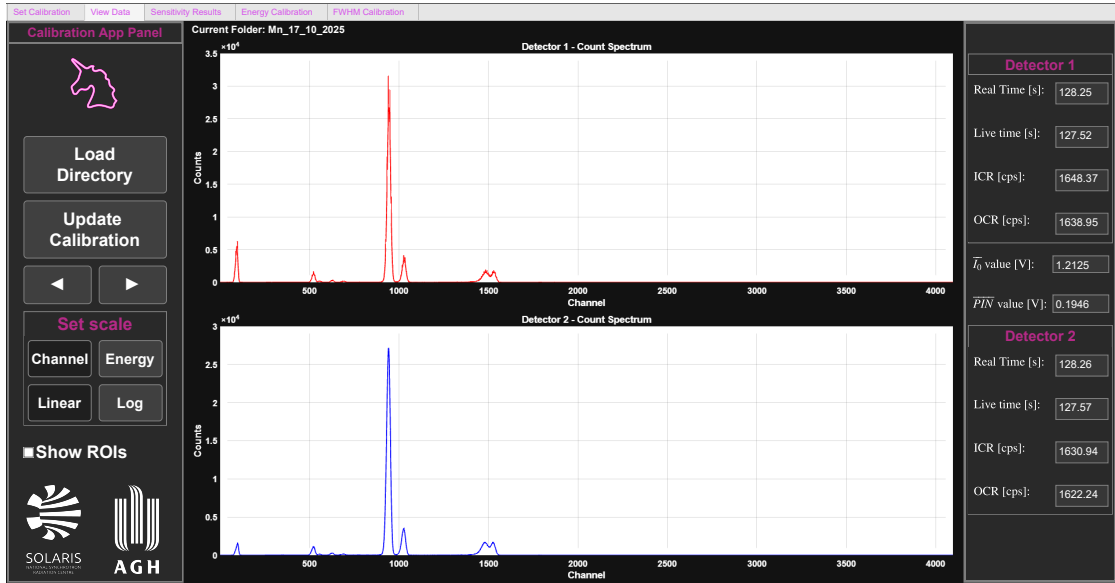


Figure 9: The basic spectrum analysis view for a manganese (Mn) standard. The central section displays separate plots for Detector 1 and Detector 2 in channel scale, while the side panel on the right provides detailed acquisition statistics.

The control panel on the left side manages data navigation and visualization settings. Navigation buttons allow user to switch between individual measurements of the standards.

Below the navigation controls, a set of switch buttons enables selection of the data presentation mode. The data can be displayed either as raw spectra expressed in detector channels or converted to a calibrated energy scale. It is also possible to toggle between **Linear** and **Log** views, which changes the Y-axis scaling from linear to logarithmic.

The panel on the right side presents detailed acquisition statistics (Real Time, Live Time, ICR, OCR) for the currently displayed measurement, visually aligned with the sum

spectrum of the corresponding detector. Additionally, the beam monitors are displayed, including the I_0 and PIN diode voltage ($\overline{PIN}$) for the measurement of the given standard.

Sum spectra and statistical informations are dynamically updated as new measurement data becomes available, according to the active display settings.

An important feature of this panel is the Show ROIs checkbox. A region of Interest (ROI) defines the energy range from which the program sums photon counts to calculate the intensity of a specific fluorescence line. Enabling this option automatically switches the plot axes to an energy scale [keV]. Colored regions under the peaks then appear on the plots, representing ROI ranges for different emission lines: K_α in black, K_β in green, L_α in blue, L_β in pink and M_α in orange. Each region indicates the spectrum region included in the intensity and sensitivity calculations, with the vertical dashed line marking the peak center determined by the algorithm (Figure 10). The ROI width is dynamically calculated for each peak based on the peak width settings defined in the “Set Calibration” tab (ROI width function). This allows verification that the ROI fully captures the signal without including excessive background or overlapping neighboring lines. The ROI width at a given energy E is calculated according to the relation:

$$\sigma^2 = \left(\frac{\text{NOISE}}{2\sqrt{2 \ln(2)}} \right)^2 + \varepsilon \cdot \text{FANO} \cdot E. \quad (6)$$

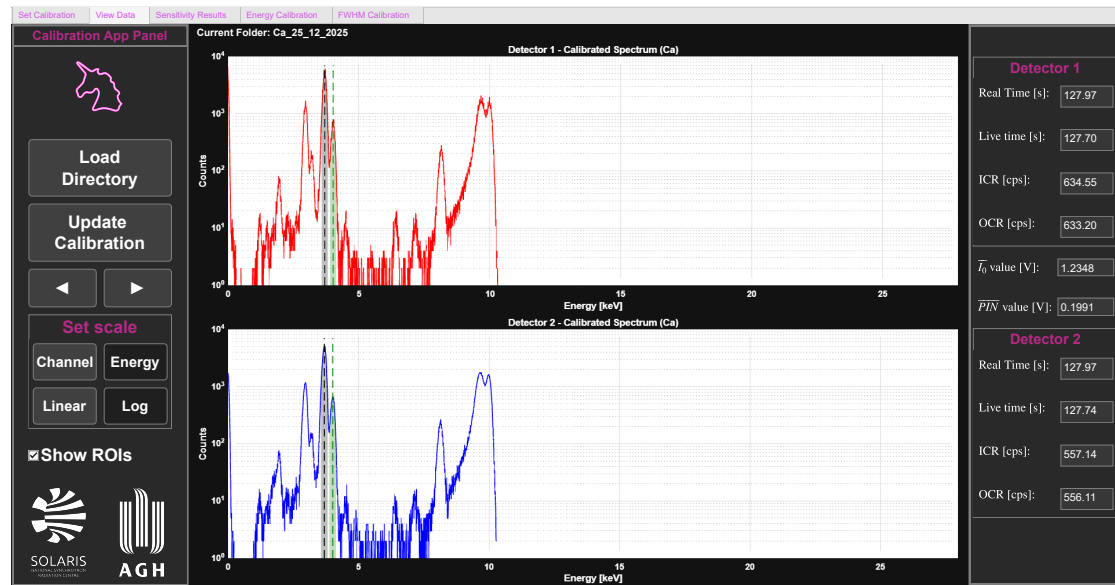


Figure 10: Visualization of the calcium (Ca) spectrum after switching the display mode to energy and logarithmic scale. The Show ROIs checkbox is selected, and the defined ROI ranges within $\pm 2\sigma$ for K_α (black) and K_β (green) lines are displayed on the plots.

By navigating to the end of the list (the last page), the user can access the combined view (Figure 11). In this view, the spectra of all loaded elements are overlaid on a single plot, enabling direct comparison for their intensities and peak positions. In this view, acquisition statistics normally available for individual spectra are not shown in the right panel, as their cumulative presentation over the entire series would not be representative.

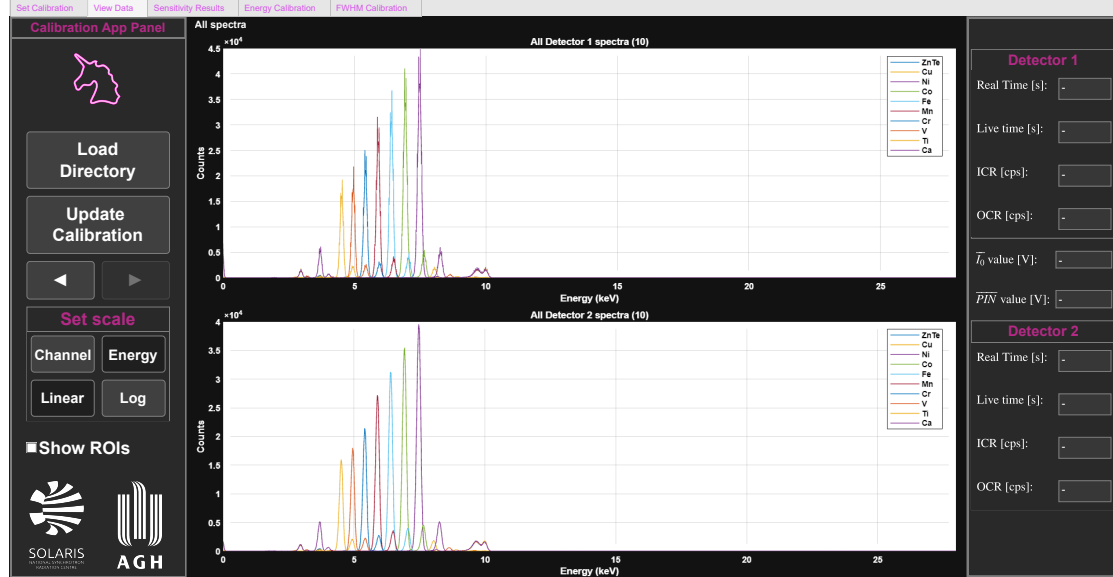


Figure 11: The overlay view displays the spectra of all ten loaded standards overlaid on each other.

3.2 Offline data analysis

The software includes offline operation capabilities, allowing previously acquired measurement data to be loaded and analyzed without requiring any connection to the experimental setup.

This enables repeatable data processing, comparison of measurement series, and recalculation of results without performing new measurements. This solution also facilitates work outside the measurement station and ensures a consistent workflow for both live and file-based data.

The Load Directory button initiates multiple file processing workflow for calibration standards, while the Update Calibration button processes a single standard. The detailed procedure is described in Section 4.2.1.

4 Energy calibration

4.1 Methodology and calibration determination procedure

In the application, the energy calibration is based on the analysis of measured spectra for standards, the reference energy database, peak detection, and fitting calibration line in order to convert the spectra bins to an energy scale. The entire process is carried out separately for two detectors. Therefore, two independent calibration functions are obtained.

The basis of the fitting procedure in the software is based on defining, for each analyzed standard i , a pair of calibration points $(CH_i, E_{\text{ref},i})$. CH_i is the channel number determined by the program corresponding to the position of the peak center in the spectrum, and the $E_{\text{ref},i}$ is the corresponding theoretical energy of the K_α line for a given element.

The application builds a database of K_α line reference energies for the list of elements that are available for measurement on the holder, using the `xraylib` library.

4.2 Peak detection algorithms

In order to determine the fitting parameters, an accurate determination of CH_i values is required. The program implements two different peak detection mechanisms, depending on the data processing mode.

4.2.1 Offline data analysis and interactive peak selection

The data processing workflow begins by clicking the Load Directory button in the “View Data” tab. This action opens a system file dialog in which the user selects a parent directory containing sub-folders with measurement data for individual calibration standards. After selecting the directory, a confirmation dialog is displayed (Figure 12), allowing verification that the program has correctly identified the total number of data sets. Once confirmed, the application automatically indexes the directory contents and starts iterative processing of each detected folder.

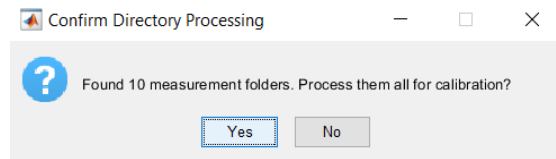


Figure 12: Measurement file import initialization window. The dialog requires confirmation of the number of detected folders before initiating sequential data processing.

For each processed folder, the first step is displaying the Standard Selection dialog (Figure 13), which associates the loaded measurement file with a specific element. The user is presented with the folder name and selects corresponding element symbol from the list of available calibration standards. This enables the software to have a prior knowledge of the expected K_{α} peak energy.

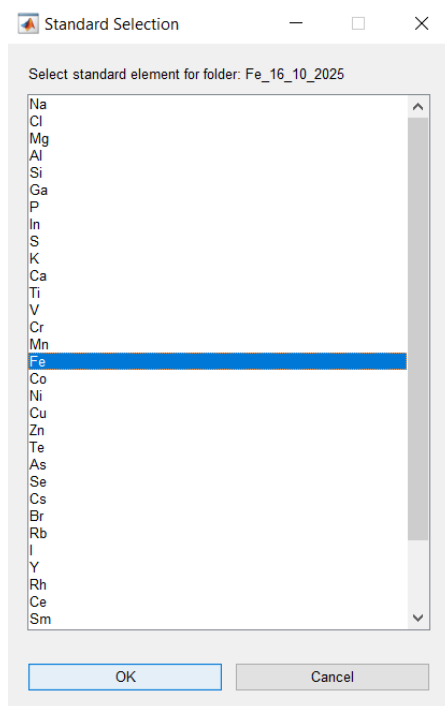


Figure 13: Calibration standard selection panel. It allows a specific element from the `xraylib` database to be assigned to the currently loaded measurement dataset.

Immediately after the standard is selected, the peak detection procedure is initiated and performed sequentially - first for Detector 1 and then for Detector 2. This opens a dedicated interactive graphical interface, the Peak Selection Window (Figure 14). Before the peak detection, the code ignores the first 300 channels of the spectrum in order to exclude the zero peak, which is a measurement artifact and typically is the highest peak in the spectrum, making further analysis difficult. Both the maximum search and subsequent analysis are therefore performed on the remaining part of the data. By default, the window starts in a zoomed-in view centered on the theoretical position of the K_α line of the selected element, within the range of ± 70 channels around the detected maximum.

The plot presents key information: the solid line represents the raw measured spectrum, the dashed line shows a Gaussian curve automatically fitted to the peak apex, and point markers indicate the peak maximum (green circle), the centroid (magenta circle), and the final determined peak center based on the fit.

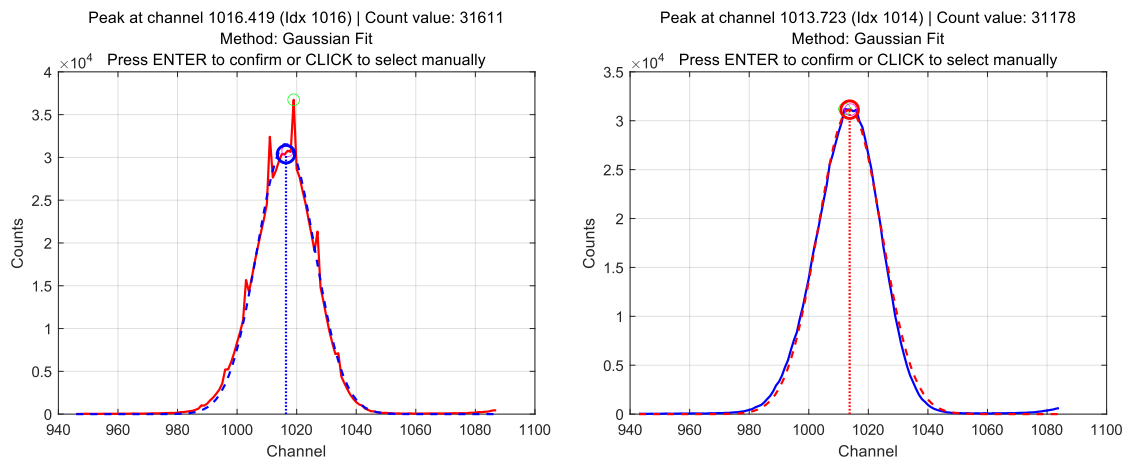


Figure 14: Default detailed view centered on the theoretical K_α line of Fe (± 70 channels around detected peak).

As an initial peak candidate, the global maximum is identified in the spectrum range starting from the first non-ignored channel to the end of the spectrum. Since the K_α line typically produces the strongest signal in this energy range, it can in most cases be effectively localized as the global maximum of the spectrum.

Around the maximum, a local window of fixed width (by default ± 15 channels) is extracted. Within this window, a Gaussian function is fitted to the data using the least squares method, with the form:

$$y(x) = a \cdot \exp\left(-\frac{(x - x_p)^2}{2\sigma^2}\right), \quad (7)$$

where:

- a — peak amplitude,
- x_p — peak position expressed in channels,
- σ — peak width expressed in channels.

Additionally, appropriate constraints are applied to the fitting parameters to ensure that the peak center remains within the defined window, the amplitude is non-negative, and the width parameter is positive.

The fit quality is verified based on the coefficient of determination R^2 . If the fit is satisfactory ($R^2 > 0.7$), the peak channel is taken as the fitted parameter x_p .

If the quality of determination using Gaussian fit is too low, a fallback procedure is applied and the peak channel is calculated as a centroid (weighted center of mass based on counts) within the local window. If this also fails, the channel with the maximum number of counts is selected.

Since this detection step in offline mode relies on a channel-based global maximum search rather than the robust algorithms in the automatic mode, the procedure may result in occasional misidentifications. Therefore, the user retains full control over the verification process to correct such errors. A right mouse button click zooms out the view to display the full spectral range, enabling a global assessment of the spectrum. This is illustrated in Figure 15 for a KCl sample, where the automatic algorithm initially failed to locate the correct spectral line. In such cases, a subsequent left mouse button click on the true peak apex activates the Manual Fit mode, which recalculates the fit parameters for the correctly selected region.

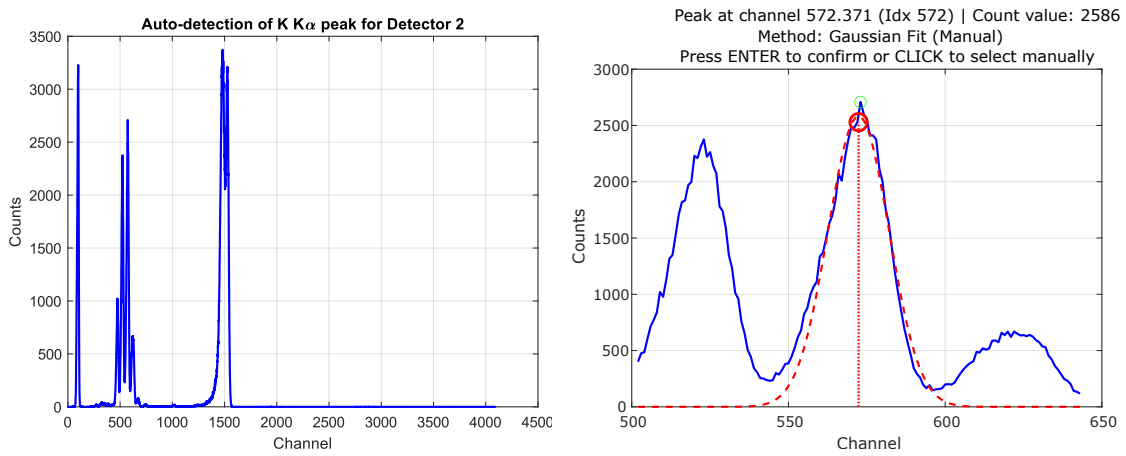


Figure 15: The Peak Selection window interface for the potassium chloride (KCl) in the full spectral view (unzoomed), activated by a right mouse button click (left). This mode allows the user to perform global assessment of the spectrum and manually correct the peak position in case of preliminary calibration errors. A subsequent left mouse button click on the selected peak recenters the analysis window and restores the detailed view (right).

Confirming the correctness of the peak detection by pressing **ENTER** key closes the window for the current detector, saves the information about the peak channel number and advances the processing loop to the next step, continuing until all folders in a selected directory have been processed.

The procedure initiated by the Update Calibration button is intended for handling individual calibration standards. Its operation is analogous to the full directory loading mode triggered by Load Directory button, with the key difference occurring at the stage of source data selection.

In this mode, the system file dialog is used to select a single, specific folder containing the measurement of a given calibration standard. After the files are loaded, the subsequent sequence of operations is identical: the application enforces element selection in the Standard Selection dialog and then launches the interactive Peak Selection window (Figure 14) for both detectors. This functionality enables rapid recalibration or the addition of a missing calibration point without the need to reprocess the entire data series.

4.2.2 Automatic mode during live measurement

In live measurement mode (on the beamline), the application automatically acquires each subsequent spectrum without needing additional user interaction, since it already knows which element of standard is being measured. This information originates from the measurement plan and the holder configuration, where, within the application, each sample slot is mapped to a list of element symbols. The element symbol is explicitly passed to the processing function (`processAutoMeasurement()`).

The peak detection process takes place in two stages based on the calibration progress, to ensure accurate results.

When the program does not yet have enough measurement points to determine the calibration (less than two standards with measured K_α line), the `autoFindPeakChannel()` function is used. The algorithm for this function works in a similar way to the offline mode: the first 300 channels are ignored, and then searches for the global maximum. A Gaussian function is fitted within the ± 20 channels range around the detected peak to determine the CH_i value. If the fit quality, expressed by the R^2 is less than 0.7, the peak position is taken as the centroid, with the alternative of using the maximum in case of an error. Consequently, due to the logic of this initialization algorithm it is recommended to perform the calibration using at least two elements with an atomic number $Z \geq 20$. This ensures that the K_α line appears well above the ignored channels threshold and is distinct enough to be identified as the global maximum. Furthermore, Ga and As are not recommended as the highest-Z calibration standards. Due to the algorithm's reliance on the maximum, a stronger signal from a coexisting element creates a risk of misidentification, leading to wrong channel assignment.

Once at least two calibration standards have been measured, the software calculates the initial calibration parameters ($GAIN_{current}$ and $ZERO_{current}$). From this moment on, the peak detection is performed using the `autoFindPeakEnergy()` function, which relies on the peak position prediction to significantly improve the accuracy and selectivity of the algorithm. First, the spectrum expressed in channel space is transformed to the energy scale using the current calibration parameters, and then peak detection is performed directly in the energy domain (keV) for each subsequent standard.

A search range is defined around the theoretical reference energy E_{ref} . The width of this window (ΔE) is variable and depends on the energy value (a narrower range for low energies and a wider one for high energies) according to the Table below. The narrowing of the search window for lower energies is motivated by the increasing accuracy of the peak position prediction (resulting from the iterative calibration procedure) and the necessity to match the window size to the decreasing FWHM of the detector to ensure proper peak separation.

Table 2: Parameters for the dynamic search window width ΔE as a function of expected energy of the K_α peak.

E_{ref} [keV]	ΔE [keV]
$(E \geq 8.0)$	1.2
$(5.0 \leq E < 8.0)$	0.72
$(4.5 \leq E < 5.0)$	0.36
$(3.0 \leq E < 4.5)$	0.2
$(E < 3.0)$	0.18

Within the defined search window, the algorithm identifies all local maxima. To distinguish the true signal from noise or other spectral lines, each peak candidate is evaluated based on two criteria: its amplitude and its distance from the expected energy. In practice, this approach has proven to be highly effective on the tested sets of calibration standards data. In the decision-making process, position plays the primary role. The algorithm favors the peaks closest to the reference energy, treating amplitude as a secondary criterion. Ultimately, the peak selected represents the best compromise between signal strength and energy accuracy.

Around the located peak to determine its position (as in the previous methods), a narrow fitting window is extracted, with a width that also depends on energy (ranging from 0.16 keV to 0.30 keV). A Gaussian function is fitted within this window. The fit is accepted if $R^2 > 0.7$ (for $E > 3$ keV) or $R^2 > 0.5$ (for $E < 3$ keV).

If resulting peak energy E_{found} deviates from E_{ref} , checking whether a weaker but correct signal was missed.

Since the goal of calibration is to determine the channel number rather than the energy, the identified peak E_{found} is converted back to the channel scale by inverting the current calibration equation:

$$CH_i = \frac{E_{\text{found}} - \text{ZERO}_{\text{current}}}{\text{GAIN}_{\text{current}}}. \quad (8)$$

The resulting CH_i value is then added to the calibration point database, enabling model recalibration and refinement of GAIN and ZERO parameters for subsequent measurements. This approach makes each new standard measurement significantly more robust to errors by refining the calibration at every iteration.

A simplified workflow scheme of the automated peak detection process is presented in Figure 16.

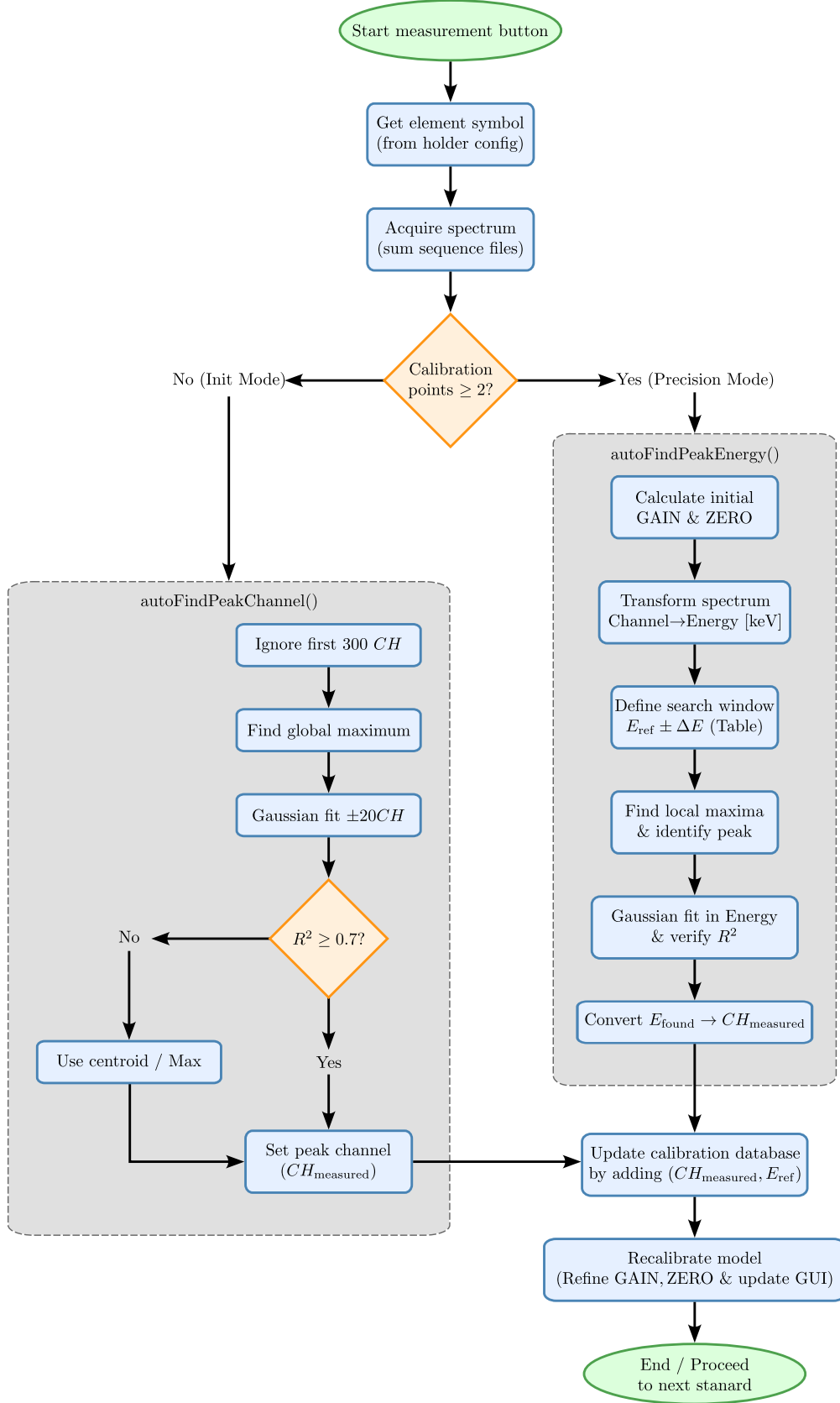


Figure 16: Simplified schematic representation of the automated peak detection and energy calibration workflow during live measurements.

4.3 Energy Calibration tab

For the calibration procedure, a linear model describing the relationship between energy and channel is used, as given by Equation:

$$E_{CH} = \text{GAIN} \cdot CH + \text{ZERO}. \quad (9)$$

To the measured data on the $E(CH)$ plot, a first-degree polynomial is fitted using the least squares method.

The results of the energy calibration process are presented in the “Energy Calibration” tab (Figure 17). This panel allows the user to view the determined calibration function and provides data and plots for evaluating quantitative and qualitative aspects of the fit for each detector.

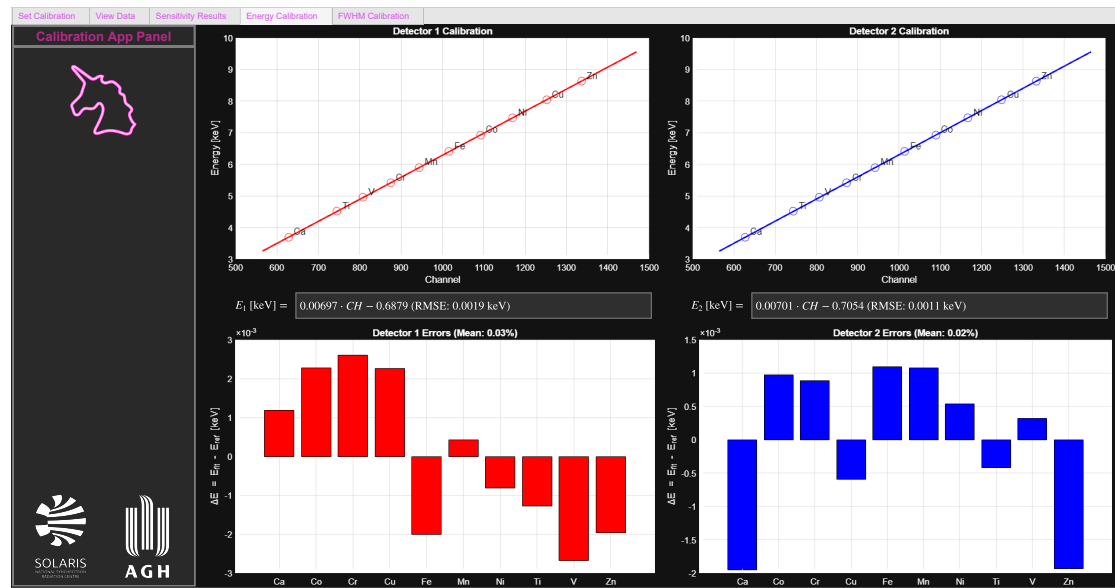


Figure 17: View of the “Energy Calibration” tab in the application interface. The plots on the left side of the panel refer to Detector 1, while the plots on the right refer to Detector 2. The upper part of the panel shows calibration plots together with determined equations and the RMSE parameter. The bar charts at the bottom illustrate the deviations between the reference and fitted values for individual measurement points.

In the upper part of the panel, graphs are displayed with measurement points plotted and a fitted line. Below, the calibration function for each detector is shown separately.

Throughout the automatic measurement process, all plots in the panel are updated after each new standard is added. The calibration coefficients GAIN and ZERO in the “Set Calibration” tab are also continuously updated. This enables the user to consistently monitor the stability and correctness of the calibration. Once the entire measurement series has been completed, the user can check the fit on the full dataset.

4.3.1 Error analysis and fit quality verification

The measure of the calibration quality is the Root Mean Square Error, determined from the formula:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (E_{\text{fit},i} - E_{\text{ref},i})^2}, \quad (10)$$

where:

- N – number of standards used for calibration,
- $E_{\text{fit},i}$ – energy of the K_α line calculated from the new calibration function for i -th channel.

In the lower part of the panel, bar charts are displayed showing the differences between the K_α line energies for the individual standards determined from the calibration and the theoretical energies.

The deviation between reference value and the fitted value for each standard is determined as:

$$\Delta E_i = E_{\text{fit},i} - E_{\text{ref},i}. \quad (11)$$

Next to each bar chart, the mean relative error is displayed:

$$\text{Mean error } [\%] = \frac{1}{N} \sum_{i=1}^N \left| \frac{\Delta E_i}{E_{\text{ref},i}} \right| \cdot 100\%. \quad (12)$$

The implemented uncertainty estimation approach, based on the analysis of energy differences (ΔE) and the RMSE metric, prioritizes diagnostic usefulness, enabling rapid verification of calibration quality. The presentation of fitting parameter using uncertainties was not included, in favor of indicators that directly reflect the accuracy of the calibration results. The RMSE provides a compact measure of the typical calibration error, expressed explicitly in energy units [keV], which makes its interpretation straightforward in the context of the measurement quality. Furthermore, presenting ΔE values in a bar-chart form enables immediate identification of a systematic trends or outliers that could remain unnoticed when considering only standard deviations of the fit parameters. It should also be noted that the calibration procedure can be performed using a minimal number of standards ($N = 2$), for which classical uncertainty estimation methods are not applicable.

5 FWHM calibration

The procedure for FWHM calibration in the application is carried out in the parallel with determining the energy calibration. The input data are obtained in the same peak detection process that was described in section 4.2.

5.1 Measurement data extraction

While the energy calibration used the peak position parameter (x_p in Gaussian curve fit), the FWHM calibration is based on the width parameter σ (equation (7)).

From the Gaussian fit, the standard deviation σ is obtained in channels. This value is then converted to Full Width at Half Maximum according to the relationship for the normal distribution and converted to an energy unit using the GAIN coefficient available after energy calibration:

$$\text{FWHM [eV]} = 2\sqrt{2 \ln 2} \cdot \sigma [\text{Ch}] \cdot \text{GAIN} \left[\frac{\text{eV}}{\text{Ch}} \right]. \quad (13)$$

As a result for each detector and every standard, a set of point pairs ($E_{\text{ref},i}$, $\text{FWHM}_{\text{eV},i}$) is obtained, relating the theoretical line energy with measured width.

5.2 Calibration model and linearization

The dependence of the energy resolution on energy is described by the model given by equation:

$$\text{FWHM}(E) = \sqrt{\text{NOISE}^2 + \left(2\sqrt{2 \ln(2)} \right)^2 \cdot \varepsilon \cdot \text{FANO} \cdot E}, \quad (14)$$

where:

- NOISE – electronic noise contribution resulting in peak broadening [eV],
- FANO – Fano factor F [eV].

In the program implementation, the square-root model was applied indirectly. Instead, the equation was linearized by squaring both sides of the equation. This leads to a linear relationship of the form: $y = kx + b$:

$$\text{FWHM}^2 = k \cdot E + \text{NOISE}^2, \quad (15)$$

where:

- $y = \text{FWHM}^2$ – dependent variable,
- $x = E$ – independent variable,
- $b = \text{NOISE}^2$ – intercept,
- $k = (2.3548)^2 \cdot \varepsilon \cdot \text{FANO}$ – slope ($\varepsilon = 3.85 \text{ keV}$ for silicon).

Using this procedure, the model parameters are obtained through a simple and efficient linear regression approach (a first-order polynomial fit).

5.3 NOISE and FANO parameters determination

Once the regression line coefficient (k, b) have been obtained, the corresponding NOISE, and FANO parameters can be derived.

The NOISE parameter is obtained by taking the square-root of the constant term. To avoid non-physical negative values, the algorithm constrains the input to be non-negative:

$$\text{NOISE [eV]} = \sqrt{\max(b, 0)}. \quad (16)$$

The slope k is then transformed into the FANO factor:

$$\text{FANO} = \max\left(\frac{k}{(2.3548)^2 \cdot \varepsilon}, 0\right). \quad (17)$$

5.4 FWHM Calibration tab

The last step of calibration is to visualize the results in the “FWHM Calibration” tab (Figure 18). The layout of this tab is analogous to the one with the energy calibration results. On two upper plots, for both detectors, the fitted curve is drawn, calculated on the basis of the extracted parameters according to the original (square-root) form of the formula.

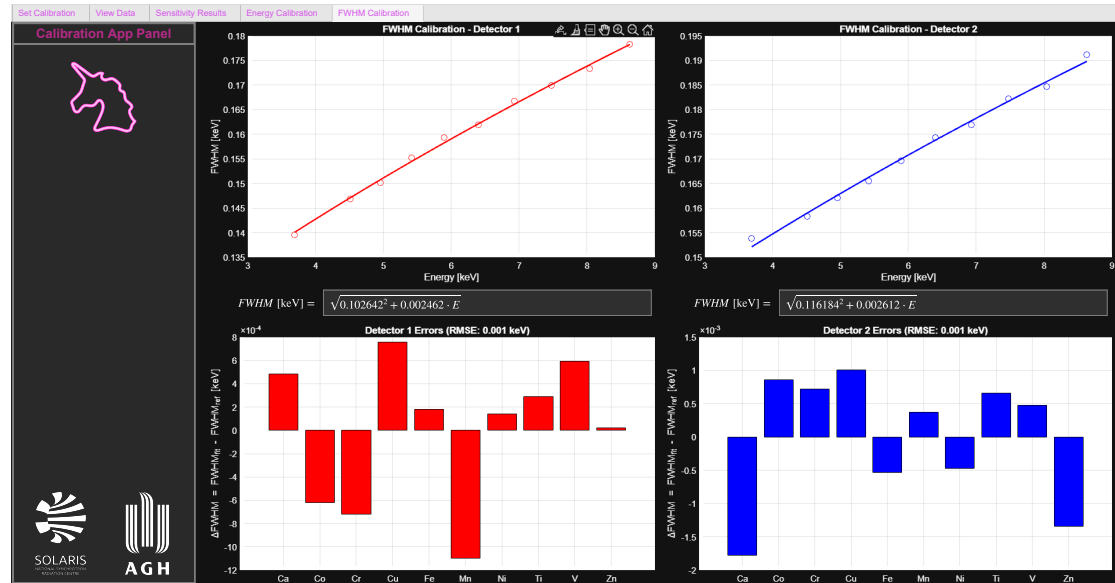


Figure 18: View of the “FWHM Calibration” tab in the application interface. The upper part of the panel shows calibration plots together with determined equations and. The bar charts at the bottom illustrate the deviations between the reference and fitted values for individual measurement points and the RMSE parameter.

In this tab, all plots are also updated in real time during measurement, and the NOISE and FANO coefficients displayed in the “Set Calibration” tab are refreshed to allow the user to continuously verify their values.

6 Sensitivity calibration

After completing the energy and FWHM calibration, the application proceeds to sensitivity calibration. This includes summing the counts within dynamically determined ROIs for the given characteristic emission line. Then, taking into account the experiment parameters (I_0 , L_T), its response relative to the mass per unit area of the measured standards is calculated.

6.1 Determination of ROI regions and sensitivity calculation

The core element of the algorithm is the method used to determine the boundaries for summing counts within a particular peak. The program uses coefficients obtained during FWHM calibration (NOISE, FANO), thereby taking into account the detector parameters at a given energy.

The model-based standard deviation of the peak in the energy scale, $\sigma(E)$ is determined according to the formula (6).

For every standard selected by the user and each emission line (for example, K_β , L_α), the detection algorithm implemented in the `autoFindPeakEnergy` function is used (discussed in detail for the K_α line in Section 4.2.2). For each line, the precise energy corresponding to the peak center is determined, around which an ROI is defined within the range $\pm n \cdot \sigma$ as:

$$\text{ROI} = [E - n \cdot \sigma(E), E + n \cdot \sigma(E)], \quad (18)$$

where:

- n – multiplier of the width around the ROI window,
- E – determined peak center energy of a given line.

The value of the parameter n is selected by the user from the `ROI width` dropdown menu (in the “Set Calibration” tab). Choosing $n = 1$ corresponds to integrating over the range $\pm 1\sigma$, which covers approximately 68% of the peak area.

The next step is the determination of the net peak area A by summing all the counts within the defined ROI interval. The program then retrieves the acquisition statistics for each measured standard: the Live Time L_T and the measured voltage of the incident X-ray beam I_0 . At the same time, the mass per unit area M assigned to the analyzed standard is obtained from a predefined parameter structure. All values corresponding to the individual MicroMatter standards are summarized in Table 1. With the complete set of data available, the software first calculates the normalized intensity as:

$$I_{E_i}^{\text{line}} = \frac{A}{I_0 \cdot L_T}, \quad (19)$$

The corresponding sensitivity is then calculated according to the formula:

$$K_{E_i}^{\text{line}} = \frac{I}{M}. \quad (20)$$

The resulting sensitivity value is expressed in units of $[\text{counts} \cdot \text{s}^{-1} \cdot \text{V}^{-1} \cdot (\mu\text{g} \cdot \text{cm}^{-2})^{-1}]$.

6.2 Results of sensitivity calibration

In the “Sensitivity Results” tab, a set of scatter plots is presented, showing the sensitivity curves. The main window is divided into four sections.

The top row displays how sensitivity varies with the atomic number (Z). The bottom row presents the same sensitivity values, but plotted as a function of the emission line energy. In both rows, the plots for Detector 1 are shown on the left, while those for Detector 2 are shown on the right, allowing direct comparison between them.

Since the sensitivity is determined for each selected by the user emission line, several views are available and can be accessed using the arrow navigation buttons located in left-hand panel. Each view corresponds to a different emission line. The default view shown in Figure 19 presents the results for the K_α line.

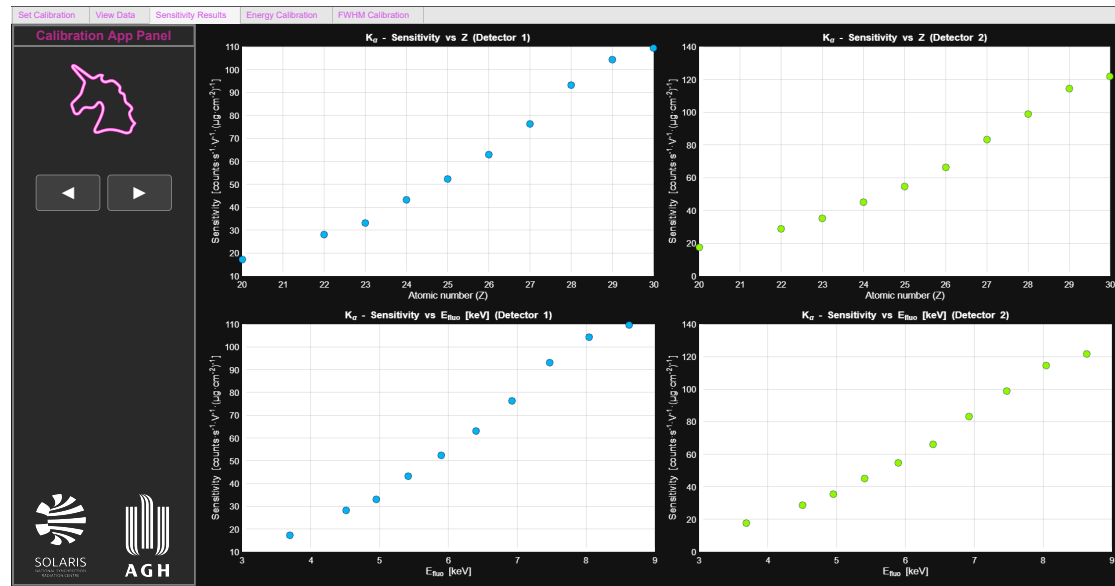


Figure 19: Results of the sensitivity calibration for the K_α lines. The upper plots present the sensitivity as a function of the atomic number (Z), while the lower plots present the sensitivity as a function of the emission line energy E_{line} for both detectors.

Subsequent views allow the analysis of other lines, such as K_β shown in figure 20. Switching between these views makes it possible to clearly observe the difference in the shape of sensitivity curves for different characteristic lines.

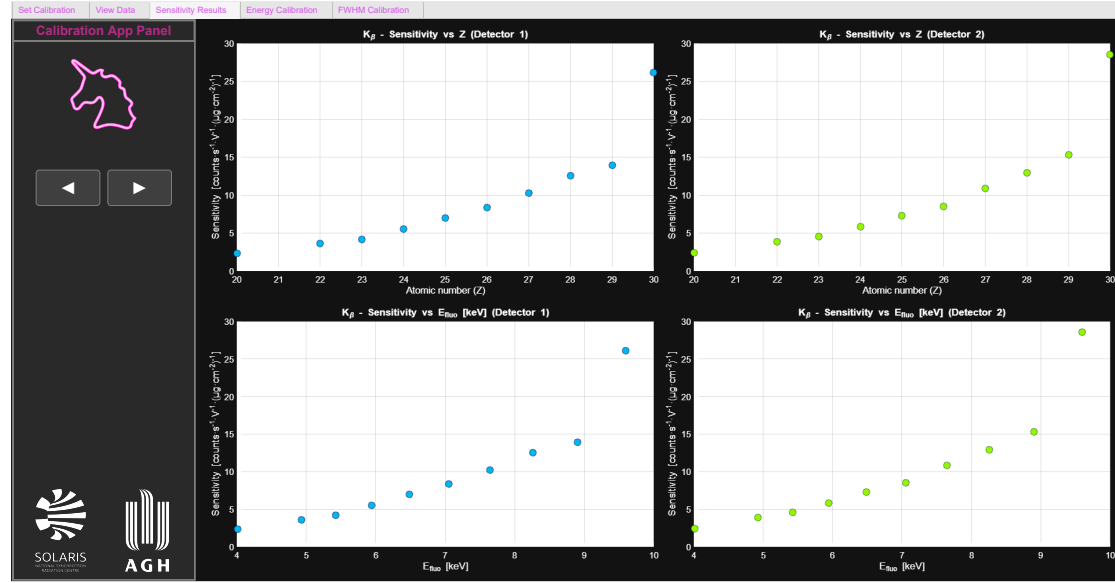


Figure 20: Results of the sensitivity calibration for the K_β lines. The upper plots present the sensitivity as a function of the atomic number (Z), while the lower plots present the sensitivity as a function of the emission line energy E_{line} for both detectors.

The correctness of the calculated sensitivity values can be verified using the spectrum visualization tools available in the “View Data” tab. When the “Show ROIs” option is enabled, the user can inspect the exact integration ranges applied by the algorithm. On the spectra, the regions of interest are highlighted within the interval $[E - n\sigma, E + n\sigma]$, as defined by Equation (18).

This verification step makes it possible to confirm that the determined ROI width fully covers the peak while avoiding excessive background contribution, thereby validating the sensitivity calibration. If the current ROI definition proves suboptimal, the calibration parameters can be easily adjusted. The user may modify the sigma multiplier n and trigger a recalculation by pressing “Update Sensitivity calibration” button.

7 Calibration results export

The application automatically archives the results in **.xlsx** (Excel) and **.txt** (text) formats, storing the measurement acquisition data and the calibration results statistics in the dedicated files.

The **MeasurementSummary_dd_mm_yyyy.xlsx** file (where the suffix corresponds to the file creation date) serves as a record of a single data processing session. It is generated and automatically overwritten after processing each data folder. The file structure includes a header section containing the current energy and FWHM calibration equations. Below the header, a data table summarizes the statistics for each processed standard, including the source folder name, atomic number Z , acquisition statistics: Real Time, Live Time, ICR, OCR and the calculated Dead Time, as well as the energies derived in the calibration process for the individual K_α lines. An example output of this file is presented in Figure 21.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Energy calibration function D1: $E = 0.006974 \cdot CH + -0.687947$													
2	Energy calibration function D2: $E = 0.007006 \cdot CH + -0.703665$													
3	FWHM calibration function D1: $FWHM = \sqrt{(0.103022^2 + 0.002450 \cdot E)}$ [keV]													
4	FWHM calibration function D2: $FWHM = \sqrt{(0.112601^2 + 0.002737 \cdot E)}$ [keV]													
5														
6														
7	Folder	Z	RealTime_D1	LiveTime_D1	ICR_D1	OCR_D1	DeadTime_D1 [%]	RealTime_D2	LiveTime_D2	ICR_D2	OCR_D2	DeadTime_D2 [%]	Energy_D1 [keV]	Energy_D2 [keV]
8	Te_25_12_2025	52	4.952579	4.927691	1342.27871	1335.50294	0.504796131	4.957506	4.933146	1357.12198	1350.42077	0.493780563	8.629006366	8.630913943
9	Zn_25_12_2025	30	4.952579	4.927691	1342.27871	1335.50294	0.504796131	4.957506	4.933146	1357.12198	1350.42077	0.493780563	8.629006366	8.630913943
10	Cu_25_12_2025	29	5.270368	5.218896	2548.77543	2523.82825	0.978791003	5.275286	5.224184	2661.06879	2635.23041	0.970977647	8.94348789	8.939629282
11	Ni_25_12_2025	28	128.460337	127.336041	2347.02279	2326.47907	0.875309441	128.465529	127.385329	2376.94841	2356.95958	0.840494469	7.471491705	7.472155847
12	Co_25_12_2025	27	127.970939	126.992267	2101.58297	2085.50806	0.764895466	127.9762	127.046792	2111.92636	2096.58743	0.726300325	6.927529138	6.925699947
13	Fe_25_12_2025	26	128.311752	127.457968	1871.74625	1859.28719	0.66563843	128.31698	127.510472	1874.25516	1862.47308	0.628627212	6.397514329	6.400261582
14	Mn_25_12_2025	25	128.251674	127.518818	1648.37055	1636.94905	0.571564537	128.256916	127.572444	1630.94236	1622.23683	0.533773086	5.895395037	5.895840751
15	Cr_25_12_2025	24	127.994958	127.402134	1365.581	1359.25421	0.463304125	128.00233	127.452961	1328.7368	1323.05452	0.427045264	5.414197381	5.412437455
16	V_25_12_2025	23	127.73592	127.226936	1197.81802	1195.04317	0.39862831	127.74113	127.274726	1148.83091	1144.63534	0.365202877	4.946947484	4.950051694
17	Ti_25_12_2025	22	128.525921	128.060785	1109.49722	1105.48014	0.362063152	128.53119	128.107966	1047.1105	1043.66149	0.329383817	4.507593103	4.508683467
18	Cs_25_12_2025	20	127.969136	127.697172	634.548838	633.199027	0.212719824	127.974412	127.737712	557.14238	556.111352	0.185056453	3.691649252	3.688999617

Figure 21: An example excerpt from the output file **MeasurementSummary.xlsx** generated after the completion of calibration process.

In addition, after completing a full measurement sequence or after a manual calibration update is triggered, a complete report is generated and saved simultaneously in **.xlsx** and **.txt** formats as **SensitivityResults_[dd_mm_yyyy_hh_mm]_E[Energy]_geom_[Geometry]_opt_[Optics].[ext]**. In this file name, fields in square brackets correspond, respectively, to the beam energy, the detector geometry, and the optics configuration. These parameters are obtained from the beamline infrastructure using local software functions (**wm_zolix**, **get_motor_pos**, and **get_corex_conf**). This file structure includes the calibration parameters values for both detectors (GAIN, ZERO, NOISE, FANO), followed by a results table. The table contains data for each analyzed characteristic line (K, L and M series) of the measured element, including the theoretical energy, the energy determined from calibration, the net peak area, the normalized intensity, the calculated sensitivity, and the Boolean flag indicating whether the Gaussian fit for a given line was successful. An example output of sensitivity calibration results file is provided in Table 3 in the Appendix.

8 Software validation and results

The developed software was validated using experimental data collected at the PolyX beamline. The correctness of all algorithms and calibration procedures was verified. Tests were carried out across various measurement configurations, including two types of monochromators (Si(111), Mo), different geometries, and both available detectors.

This section presents representative results for Detector 2 in a configuration with Si(111) monochromator, backscatter geometry, and the beam energy of 10 keV. Complete datasets for this configuration (both detectors, all spectra) are provided in Section 8.1 of the Appendix.

The automatic procedure for determining K_α peaks for ten standards (Zn, Cu, Ni, Co, Fe, Mn, Cr, V, Ti, Ca) enabled the determination of the energy calibration parameters (Figure 22).

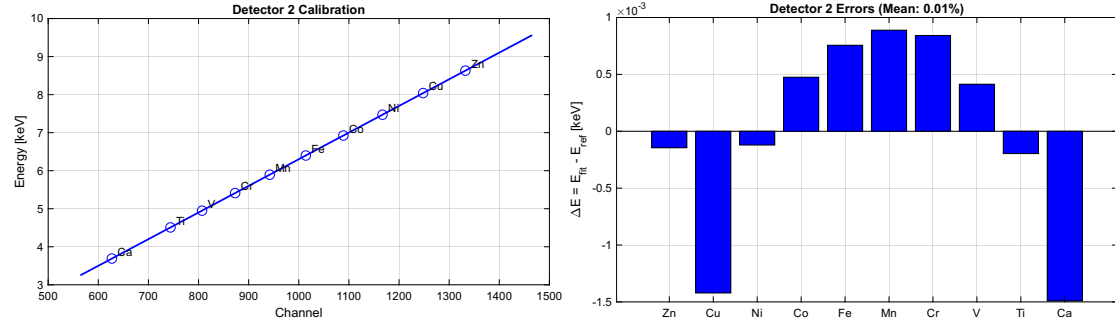


Figure 22: Energy calibration of Detector 2. The left plot shows dependence of K_α line energy on the channel number together with a linear fit. The right plot presents the differences between the fitted and reference energies for the individual standards.

A linear relationship was obtained, described by the equation:

$$E_2 \text{ [keV]} = 0.00701 \cdot CH - 0.7034.$$

The obtained RMSE value was 0.7 eV, and the average deviation between theoretical and reference values was 0.01%. The maximum deviation of individual measured standard did not exceed 1 eV, demonstrating the high accuracy of the results.

The FWHM calibration results were presented in Figure 23.

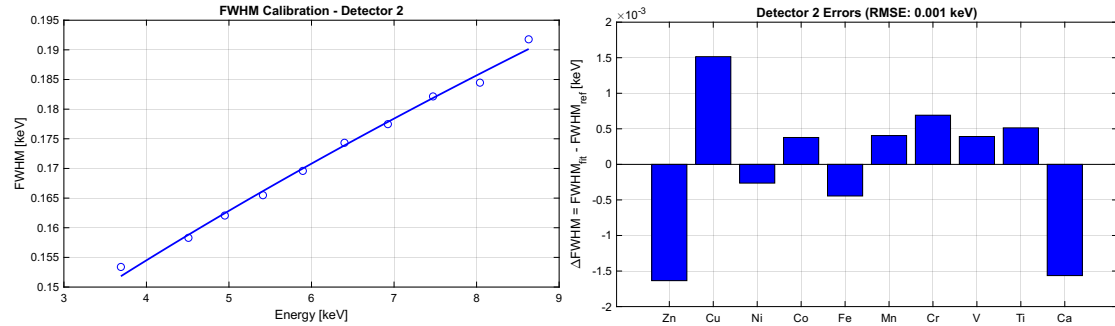


Figure 23: FWHM calibration of Detector 2. The left plot shows the square-root dependence of the FWHM on the K_α line energy. The right plot presents the differences between the fitted and reference FWHM values for the individual standards.

The dependence of FWHM on energy was successfully fitted as:

$$FWHM_2 \text{ [keV]} = \sqrt{0.115166^2 + 0.002652 \cdot E}.$$

A very low RMSE of 1 eV was achieved.

Figure 24 show sensitivity for the K_α line as a function of the atomic number and the fluorescence energy.

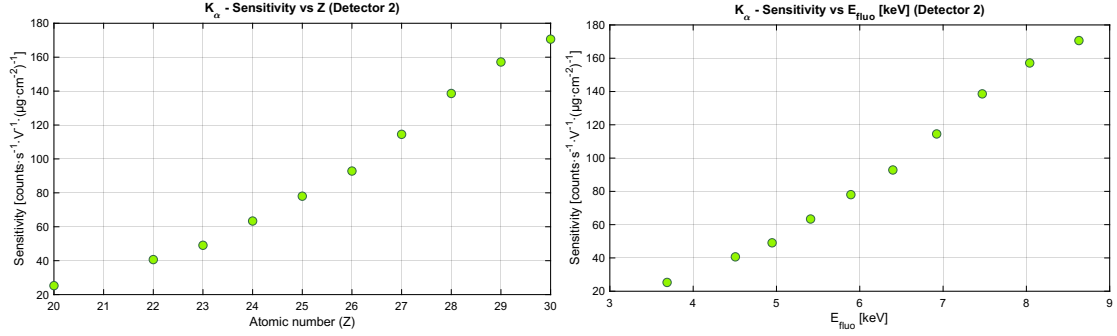


Figure 24: Elemental sensitivity curve for K_α line as a function of atomic number Z (left plot) and fluorescence energy E_{fluo} (right plot) for Detector 2.

An increase in sensitivity with increasing atomic number is observed. For the K_α line, the sensitivity values stabilizes at higher energies.

Figures 25-27 present spectra for Mn, Ti, and Te. The software correctly identified all characteristic lines for the analyzed standards (K series for Mn and Ti, and L series for Te).

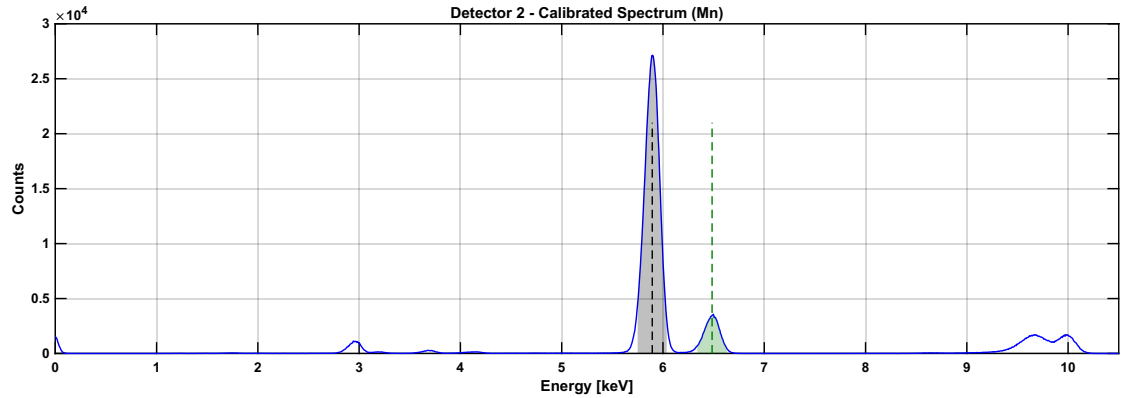


Figure 25: XRF spectrum of the Mn standard measured with Detector 2 (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

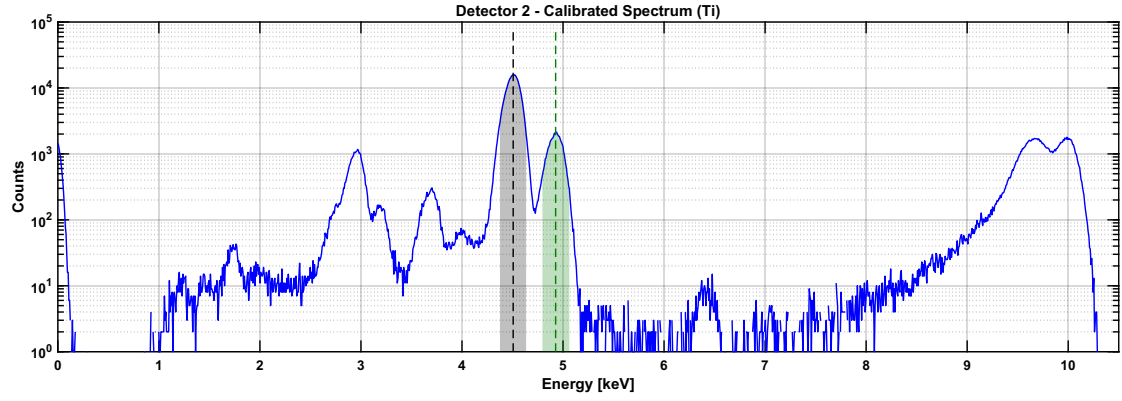


Figure 26: XRF spectrum of the Ti standard measured with Detector 1 (monochromator: Si, geometry: backscatter) in log scale. The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

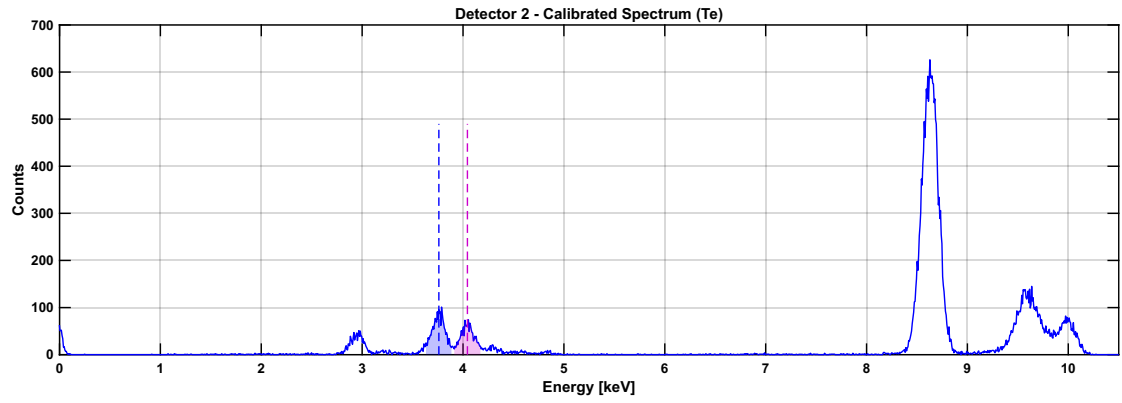


Figure 27: XRF spectrum of the Te standard measured with Detector 1 (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: blue for the L_α line and pink for the L_β line.

The overlaid spectra shown in Figure 28 illustrate the complete set of calibration data.

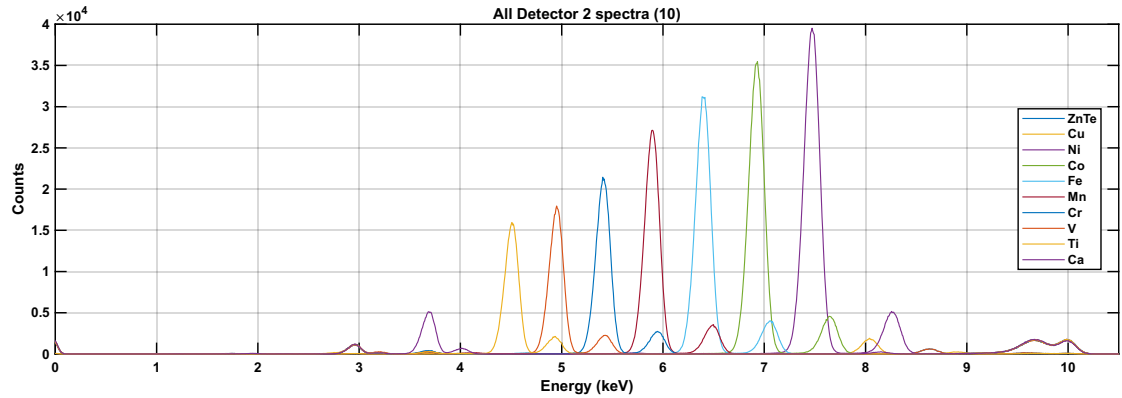


Figure 28: A set of overlaid XRF spectra of calibration standards measured with Detector 2 (Si, backscatter).

References

- [1] T. Schoonjans, A. Brunetti, B. Golosio, M. Sanchez del Rio, V. A. Solé, C. Ferrero, and L. Vincze, “The xraylib library for X-ray–matter interactions. Recent developments,” *Spectrochimica Acta Part B: Atomic Spectroscopy*, vol. 66, no. 11-12, pp. 776–784, Nov. 2011.

Appendix

8.1 Monochromator: Si, geometry: backscatter

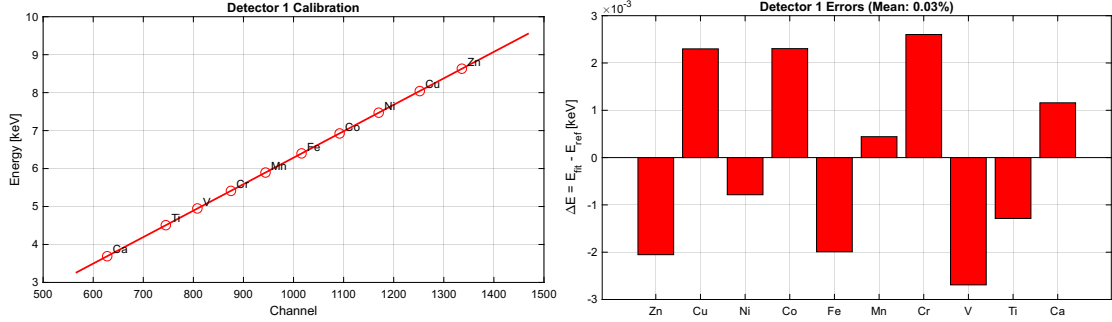


Figure 29: Energy calibration of Detector 1. The left plot shows dependence of K_{α} line energy on the channel number together with a linear fit. The right plot presents the differences between the fitted and reference energies for the individual standards.

A linear relationship for Detector 1 was obtained, described by the equation:

$$E_1 \text{ [keV]} = 0.00698 \cdot CH - 0.6892 \text{ (RMSE : 0.0017 keV)}.$$

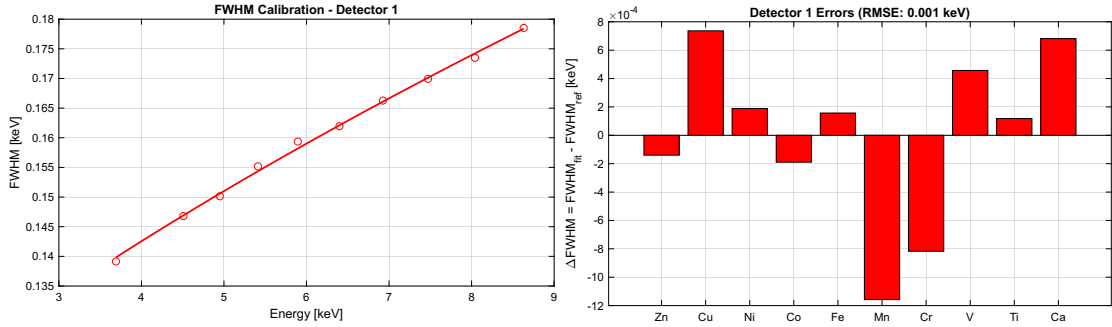


Figure 30: FWHM calibration of Detector 1. The left plot shows the square-root dependence of the FWHM on the K_{α} line energy. The right plot presents the differences between the fitted and reference FWHM values for the individual standards.

The dependence of FWHM on energy was successfully fitted for Detector 1 as:

$$FWHM_1 \text{ [keV]} = \sqrt{0.101917^2 + 0.002484 \cdot E}.$$

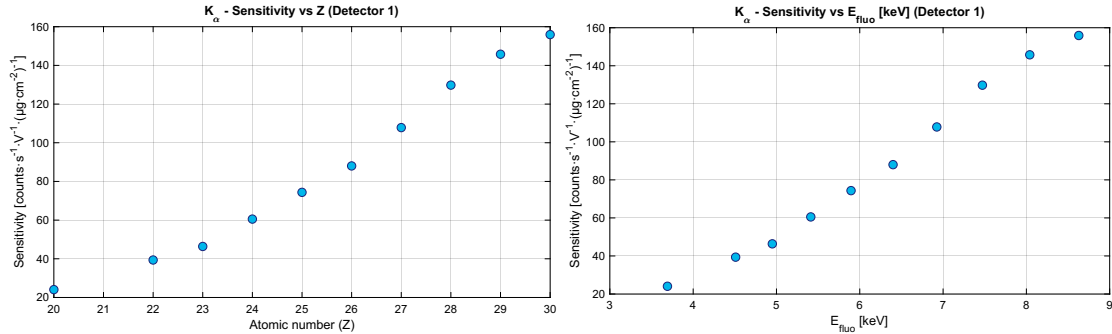


Figure 31: Elemental sensitivity curve for K_{α} line as a function of atomic number Z (left plot) and fluorescence energy E_{fluo} (right plot) for Detector 1.

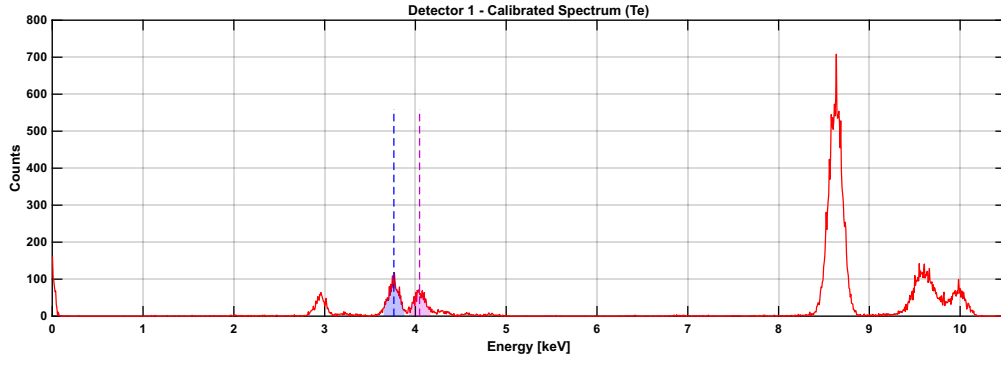


Figure 32: XRF spectrum of the Te standard measured with Detector 1 (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: blue for the L_α line and pink for the L_β line.

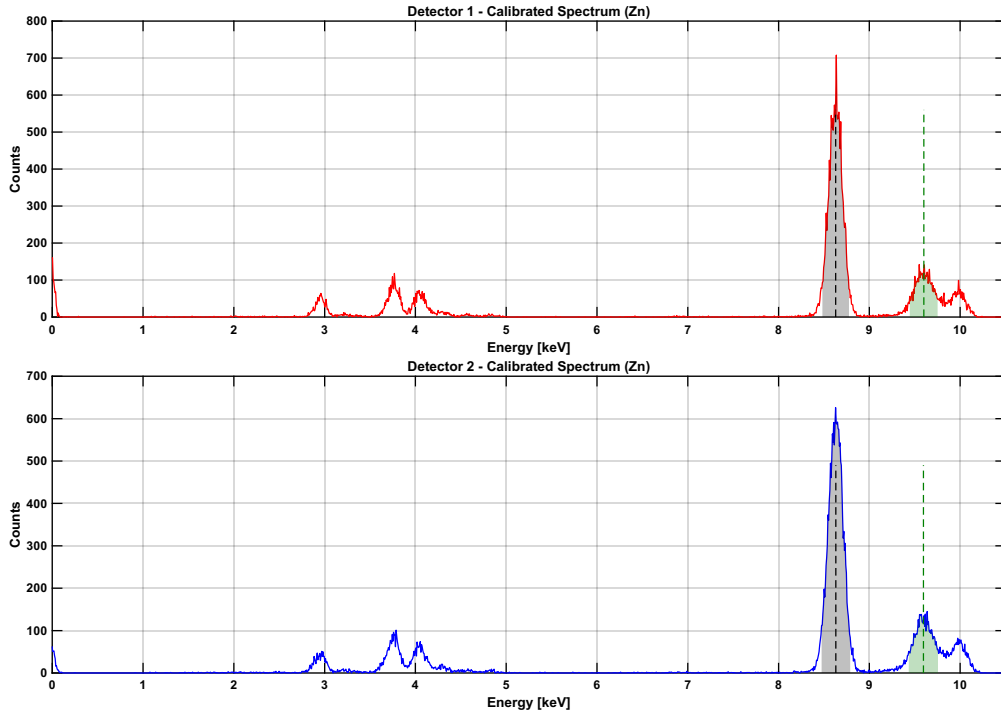


Figure 33: XRF spectrum of the Zn standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

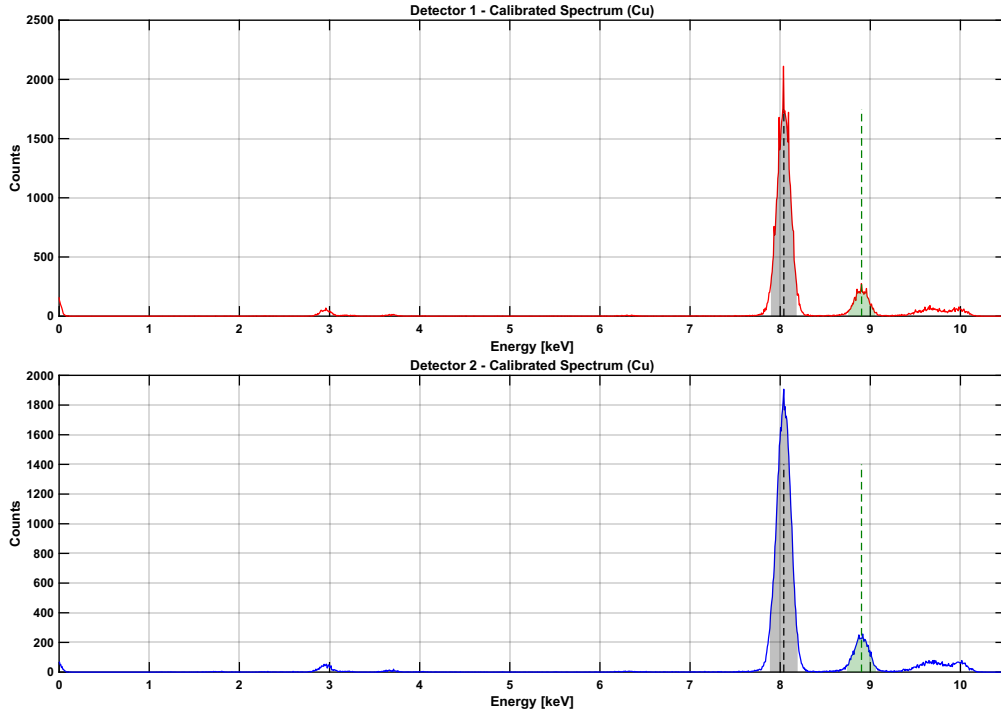


Figure 34: XRF spectrum of the Cu standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

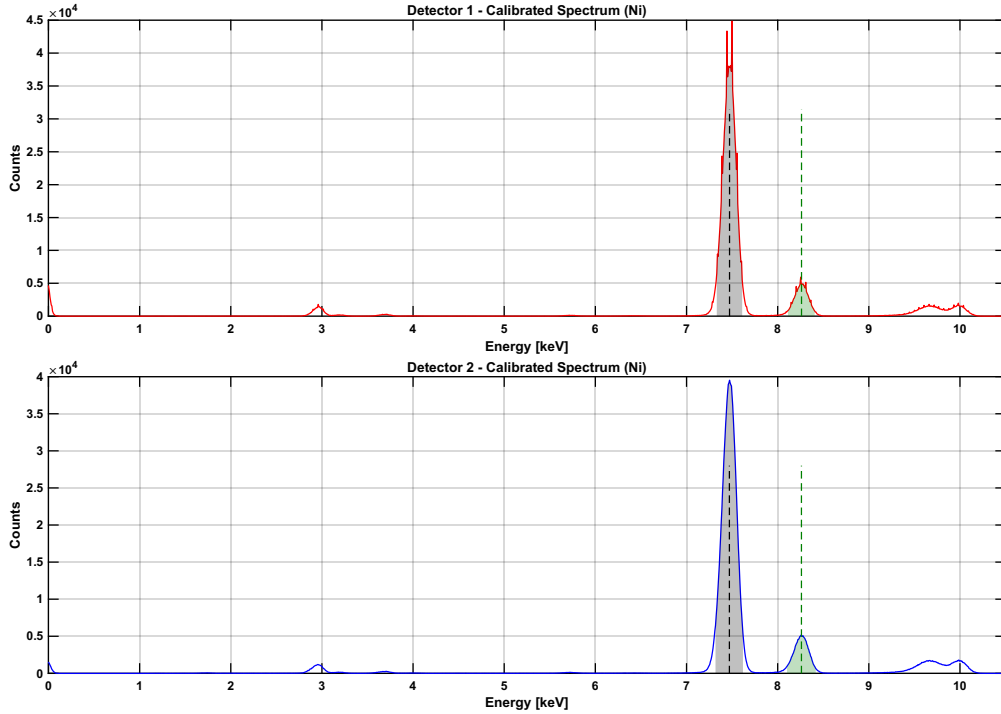


Figure 35: XRF spectrum of the Ni standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

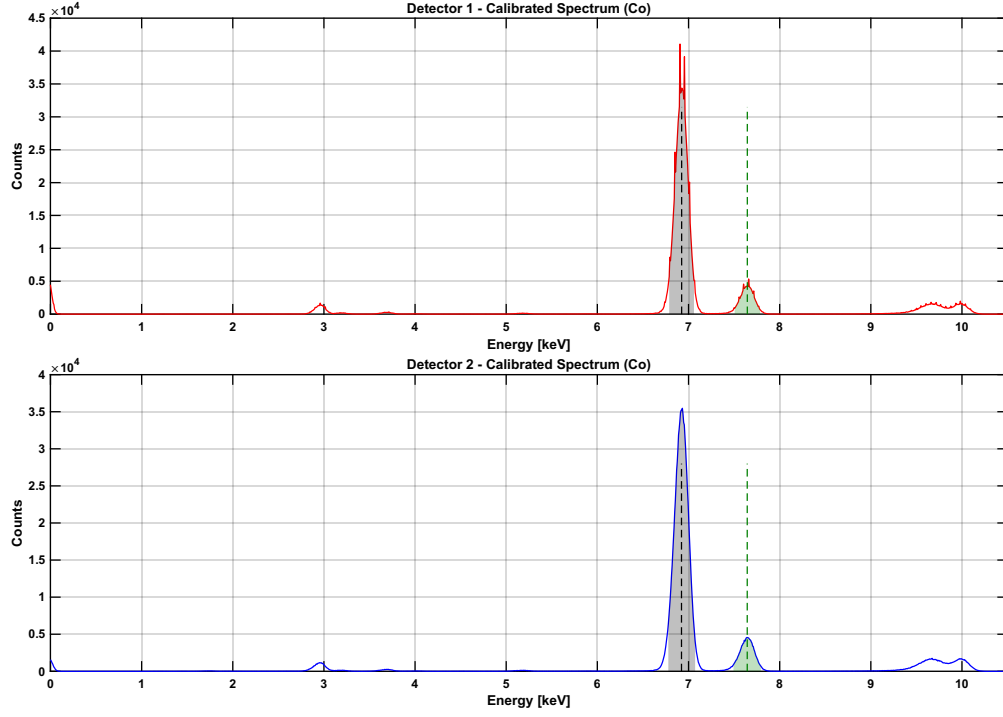


Figure 36: XRF spectrum of the Co standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

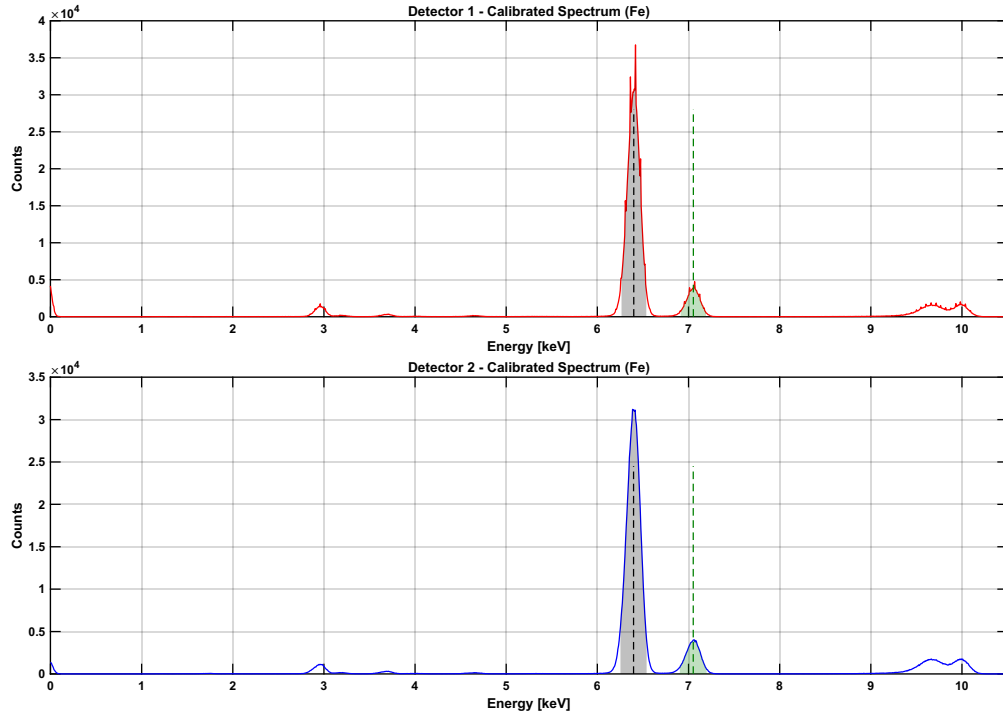


Figure 37: XRF spectrum of the Fe standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

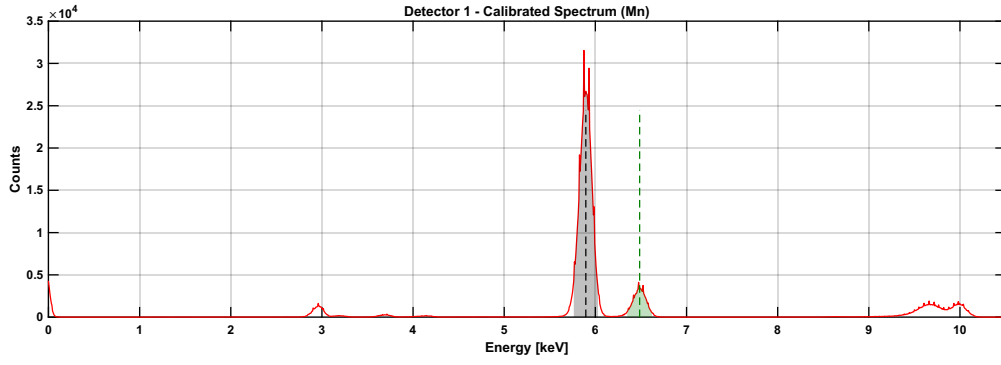


Figure 38: XRF spectrum of the Mn standard measured with Detector 1 (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

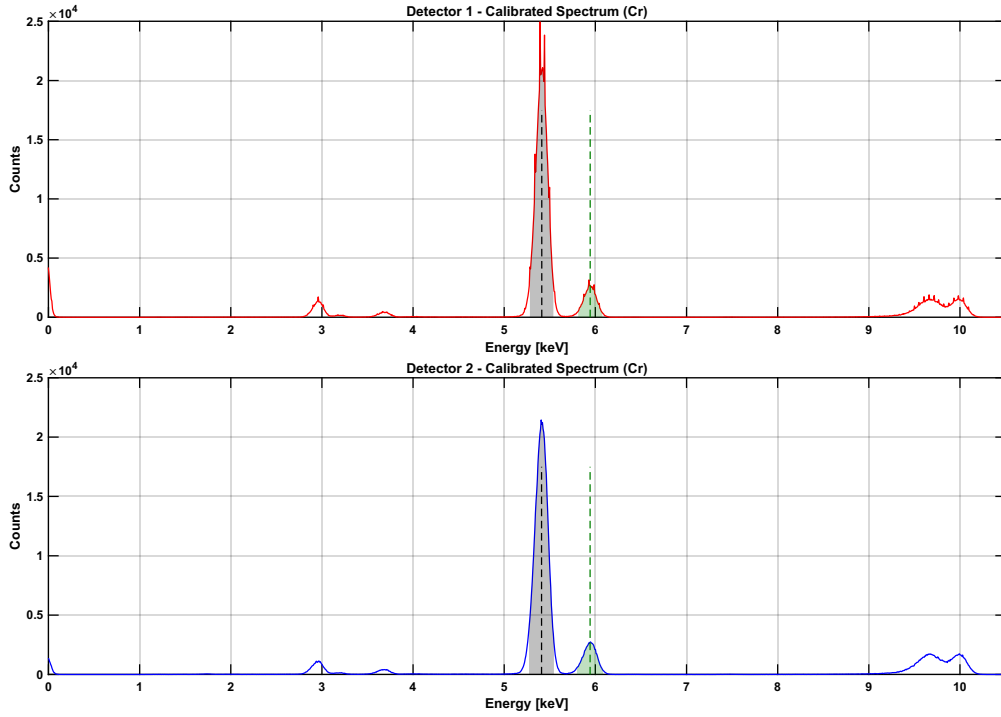


Figure 39: XRF spectrum of the Cr standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

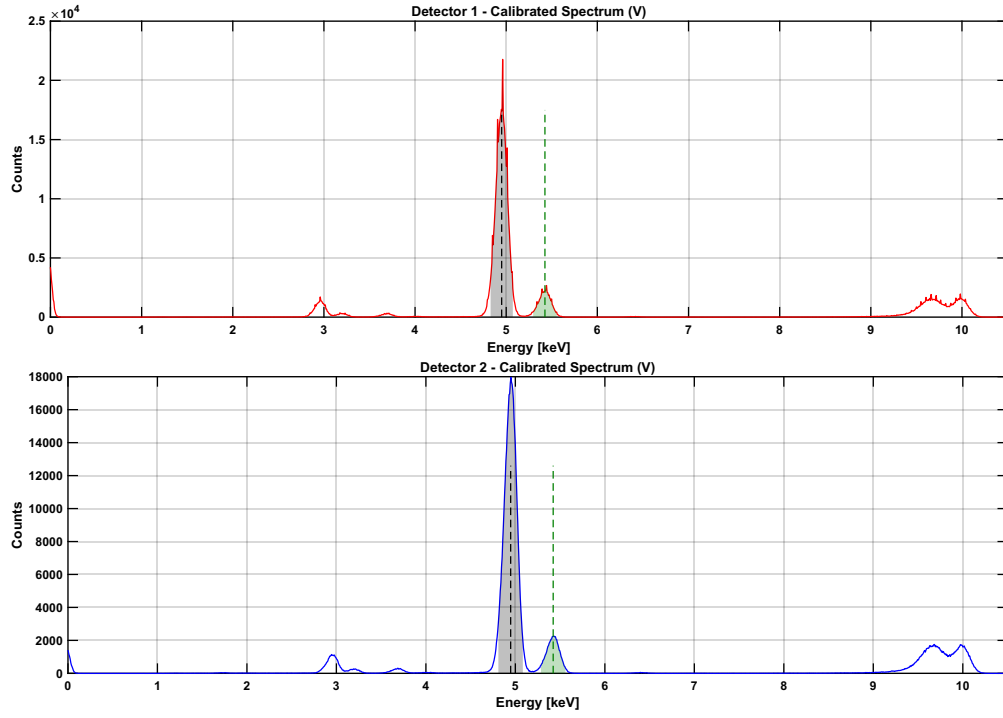


Figure 40: XRF spectrum of the V standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

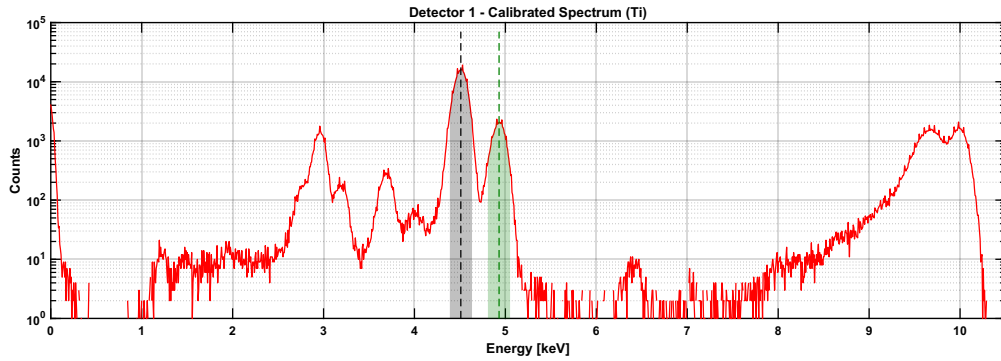


Figure 41: XRF spectrum of the Ti standard measured with Detector 1 (monochromator: Si, geometry: backscatter) in log scale. The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_α line and green for the K_β line.

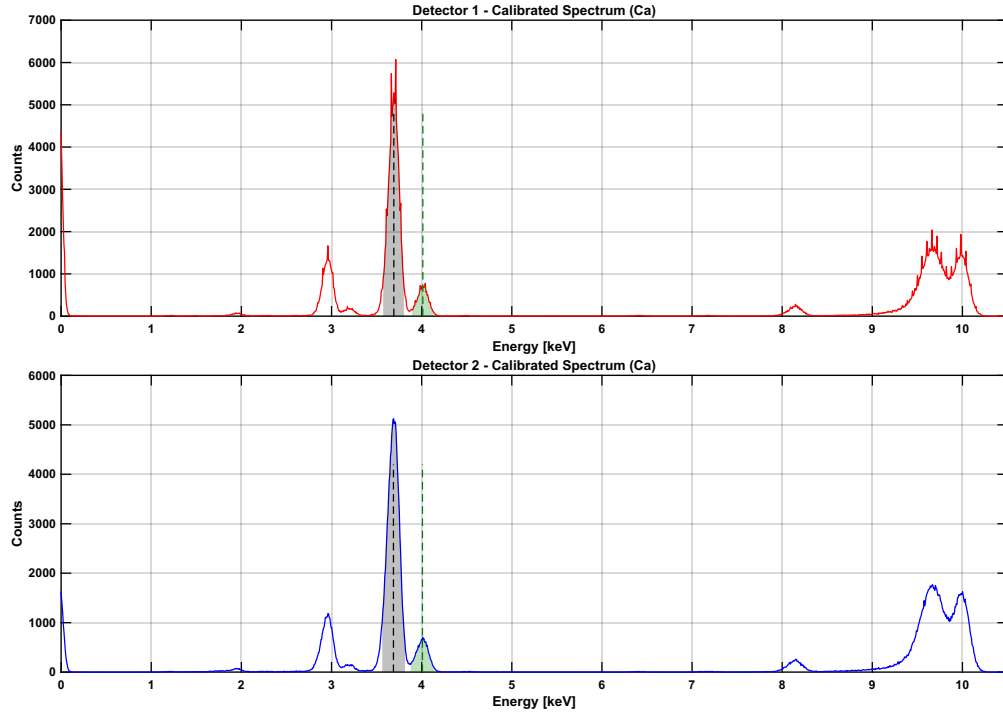


Figure 42: XRF spectrum of the Ca standard measured with Detector 1 (top) and Detector 2 (bottom) (monochromator: Si, geometry: backscatter). The ROI regions ($\pm 2\sigma$) are highlighted in color: black for the K_{α} line and green for the K_{β} line.

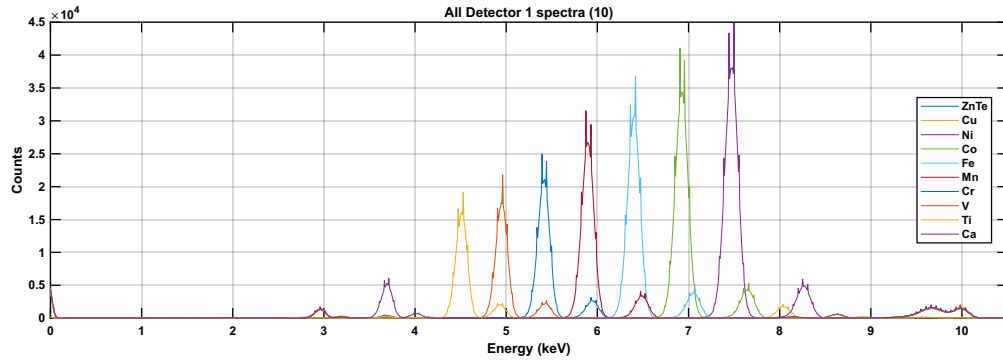


Figure 43: A set of overlaid XRF spectra of calibration standards measured with Detector 1 (Si, backscatter).

Table 3: Sensitivity Calibration Results table generated after completion of the calibration data presented in Section 8.

Calibration Parameters:	
GAIN1	6.9739 eV/ch
ZERO1	-687.9468 eV
GAIN2	7.0058 eV/ch
ZERO2	-703.6651 eV
NOISE1	101.8943 eV
FANO1	0.11636
NOISE2	115.1662 eV
FANO2	0.12423

Element	Line	Z	E_{line} [keV]	Found _{E1} [keV]	Found _{E2} [keV]	Area _{D1} [counts]	Area _{D2} [counts]	I_{D1} [counts s ⁻¹ V ⁻¹]	I_{D2} [counts s ⁻¹ V ⁻¹]	Sens _{D1} [counts s ⁻¹ V ⁻¹ (μg cm ⁻²) ⁻¹]	Sens _{D2} [counts s ⁻¹ V ⁻¹ (μg cm ⁻²) ⁻¹]	Surf_Mass [μg cm ⁻²]	I_0 [V]	LiveT _{D1} s	LiveT _{D2} s	Fit1	Fit2
Te	L _α	52	3.768	3.762	3.760	1920	1970	322.424	330.455	10.07576075	10.32671857	32	1.208	4.928	4.933	PRANDA	PRANDA
Te	L _β	52	4.089	4.047	4.043	1457	1520	244.673	254.970	7.646	7.968	32	1.208	4.928	4.933	PRANDA	PRANDA
Zn	K _α	30	8.631	8.629	8.631	15228	16682	2557.228	2798.300	155.929	170.628	16.4	1.208	4.928	4.933	PRANDA	PRANDA
Cu	K _α	29	8.041	8.041	8.040	45292	48878	7186.424	7747.560	145.769	157.151	49.3	1.208	5.219	5.224	PRANDA	PRANDA
Ni	K _α	28	7.472	7.472	7.470	976752	1043347	6074.667	6486.327	129.801	138.597	46.8	1.263	127.336	127.385	PRANDA	PRANDA
Co	K _α	27	6.925	6.925	6.923	853681	906908	5348.479	5679.518	107.832	114.506	49.6	1.257	126.992	127.047	PRANDA	PRANDA
Fe	K _α	26	6.400	6.399	6.397	746719	788142	4682.340	4940.050	88.014	92.858	53.2	1.251	127.458	127.510	PRANDA	PRANDA
Mn	K _α	25	5.895	5.895	5.893	635791	667513	4111.892	4315.235	74.356	78.033	55.3	1.213	127.519	127.572	PRANDA	PRANDA
Cr	K _α	24	5.412	5.412	5.410	487543	510757	3141.575	3289.846	60.531	63.388	51.9	1.218	127.402	127.453	PRANDA	PRANDA
V	K _α	23	4.950	4.950	4.947	395694	418942	2541.635	2689.952	46.380	49.087	54.8	1.224	127.227	127.275	PRANDA	PRANDA
Ti	K _α	22	4.509	4.510	4.507	353946	365564	2248.442	2321.390	39.377	40.655	57.1	1.229	128.061	128.108	PRANDA	PRANDA
Ca	K _α	20	3.690	3.691	3.687	108163	113643	685.976	720.502	24.069	25.281	28.5	1.235	127.697	127.738	PRANDA	PRANDA
Zn	K _β	30	9.572	9.598	9.596	4075	4457	684.312	747.633	41.736	45.587	16.4	1.208	4.928	4.933	PRANDA	PRANDA
Cu	K _β	29	8.905	8.903	8.902	6109	6833	969.307	1083.086	19.661	21.969	49.3	1.208	5.219	5.224	PRANDA	PRANDA
Ni	K _β	28	8.265	8.262	8.261	131016	141411	814.822	879.130	17.411	18.785	46.8	1.263	127.336	127.385	PRANDA	PRANDA
Co	K _β	27	7.649	7.645	7.644	113658	121670	712.090	761.959	14.357	15.362	49.6	1.257	126.992	127.047	PRANDA	PRANDA
Fe	K _β	26	7.058	7.053	7.052	98760	105181	619.280	659.271	11.641	12.392	53.2	1.251	127.458	127.510	PRANDA	PRANDA
Mn	K _β	25	6.490	6.486	6.484	84581	89479	547.016	578.450	9.892	10.460	55.3	1.213	127.519	127.572	PRANDA	PRANDA
Cr	K _β	24	5.947	5.943	5.942	64342	68519	414.600	441.339	7.988	8.504	51.9	1.218	127.402	127.453	PRANDA	PRANDA
V	K _β	23	5.427	5.425	5.422	52949	55489	340.104	356.285	6.206	6.502	54.8	1.224	127.227	127.275	PRANDA	PRANDA
Ti	K _β	22	4.932	4.930	4.927	46784	49124	297.195	311.945	5.205	5.463	57.1	1.229	128.061	128.108	PRANDA	PRANDA
Ca	K _β	20	4.013	4.013	4.009	14981	15430	95.010	97.827	3.334	3.433	28.5	1.235	127.697	127.738	PRANDA	PRANDA